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ANSWERS & EXPLANATIONS

GENERAL STUDIES (P) TEST –3796 (2023)

Q 1.D

- In Jainism, the **three jewels (also referred to as ratnatraya or tri-ratna)** are understood as **samyagdarshana ('right faith'), samyagjnana ('right knowledge), and samyakcharitra ('right action').**
- One of the three cannot exist exclusive of the others, and all are required for spiritual liberation i.e. attainment of freedom from worldly bonds.
- **Right faith is the belief in the teachings and wisdom of Mahavira. Hence statement 1 is correct.**
- **Right Knowledge is the acceptance of the theory that there is no God and that the world has been existing without a creator and that all objects possess a soul. Hence statement 2 is not correct.**
- **Right conduct refers to the observance of the five great vows: - Hence statement 3 is correct.**
 - not to injure life
 - not to lie
 - not to steal
 - not to acquire property
 - not to lead immoral life

Q 2.A

- During the Sangam Period, Land revenue was the main source of income for the chiefs.
- **The Tamil literature mentions irai and tirai as two types of contributions received by the chieftains. While irai was a regular contribution, tirai was a tribute. Hence option (a) is the correct answer.**
- The rulers were often advised to be moderate in the collection of revenue indicating that coercion and excesses were practiced by authorities.
- **The resources were redistributed amongst the masses in different forms, the most popular one being gifts. The gift of a meal or a piece of cloth was a simple form of redistribution. Fighting heroes were provided with feasts both before and after plunder and raid. Many times gifts included fine imported wine, silk clothes, and even gold ornaments.**
- Brahmanas and warrior heroes were beneficiaries of land donations and cattle by way of remuneration of their services. The acts of redistribution through gifts were made by three groups of persons with wealth and power, namely, the crowned kings (vendar), the minor chieftains (velir), and the well-to-do agricultural householders (Vellalar) of the agrarian settlements. The recipients of land also received the right to collect income from land.

Q 3.D

- **Maharaja Sawai Jai Singh II (1686-1743) was the ruler of the Rajput State of Amber in India. A feudatory of the Mughals, he received the title of 'Sawai' (one and a quarter) from Emperor Aurangzeb, who declared him a quarter superior to his famous forebearer Mirza Raja Jai Singh (d. 1667) after he captured the Fort of Vishalgarh from the Marathas in 1701.**
- **After bringing to the Emperor's notice some astronomic discrepancies that possibly affected the timings of Hindu and Muslim holy events and expressing his desire to correct these, Jai Singh also received Imperial backing for building his Astronomy Observatories at Delhi, Jaipur, Varanasi, Ujjain, and Mathura.**
 - All except the observatory at Mathura are in existence today.
- Sawai Jai Singh's ingenuity led to the invention of several outsize masonry instruments, the majority of which were used to determine the coordinates of celestial objects with reference to the local horizon.
 - During Jai Singh's lifetime, the observatories were used to make observations in order to update existing ephemerides such as the Zij-i Ulugh Begi.

- Jai Singh established communications with European astronomers through a number of Jesuits living and working in India. In addition to dispatching ambassadorial parties to Portugal, he invited French and Bavarian Jesuits to visit and make use of the observatories
- Jai Singh wrote the astronomical work 'Zij-e-Muhammad-Shahi' (Muhammad Shah's astronomical tables) in 1728. That same year he also built his new, magnificently designed capital Jaipur, about 200 km southwest of Delhi and constructed by combining the aspects of the ancient Hindu treatise on architecture, the Shilpa Shastra, and plans of many European cities of the period with Jai Singh's own ideas.
- The European travelers of the time, like the Frenchman Louis Rousselet, and the English bishop, Heber, were greatly impressed by Jai Singh's unparalleled excellence in city planning.
- **Hence option (d) is the correct answer.**

Q 4.C

- The Janapadas were the major kingdoms of Vedic India. During that period, Aryans were the most powerful tribes and were called 'Janas'. This gave rise to the term Janapada where Jana means 'people' and Pada means 'foot'.
- Socio-economic developments chiefly due to the use of iron tools in agriculture and military, along with religious and political developments led to the rise of the Mahajanapadas from small kingdoms or Janapadas. This period is also known as the era of second urbanization, the first being the Harappan civilization.
- The Buddhist Anguttara Nikaya gives a list of sixteen Mahajanapadas: Anga, Assaka (or Asmaka), Avanti, Chedi, Gandhara, Kashi, Kamboja, Kosala, Kuru, Magadha, Malla, Matsya (or Maccha), Panchala, Surasena, Vajji and Vatsa (or Vamsa).
- **The capital of Vatsa was Kausambhi, located on the banks of the confluence of Ganga and Yamuna.** This Mahajanapada followed the monarchical form of governance. The ruler Udayana who made Buddhism a state religion belonged to this Mahajanapada. **Hence, pair 1 is correctly matched.**
- **The capital of Shurasena was Mathura.** The famous ruler Awantipura, who was a disciple of Buddha, belonged to this Mahajanapadas. **Hence, pair 2 is correctly matched.**
- **The capital of Asmaka was the city variously named Podana, Potali, and Potana.** It was the only Mahajanapada situated to the south of the Vindhya Range and was in Dakshinapatha. **Hence, pair 3 is not correctly matched.**
- **The capitals of Malla are Kusinara and Pava. Malla was a republic.** The Buddha took his last meal at Pava and went to Mahaparinirvana at Kusinara. **Hence, pair 4 is correctly matched.**

Q 5.D

- **The administrative structure of the Mauryan Empire involved a division of the empire into provinces, each under the direct governance of a prince (kumara) or a member of the royal family. Hence statement 1 is not correct.**
- The inscriptions suggest four such provinces – a southern one with its center at Survarnagiri, a northern province with a capital at Taxila, a western one with its capital at Ujjayini, and an eastern one with its capital at Tosali. Ashokan inscriptions also referred to these governors as kumara, suggesting a continuation of the tradition of appointing royal princes to these important posts.
- **Senior officers called pradeshikas were tasked with touring the empire every five years and performing an audit as well as keeping a check on the provincial administration.** In addition, there were **judicial officers, rajukas, in both urban and rural areas**, whose judicial functions often combined with the assessment of revenue. A well-organized administration was needed for a variety of tasks such as surplus production, extraction of surplus, distribution or expenditure, a strong army to conquer areas, tax collection from traders and agriculturalists, etc.
- **The Arthashastra and even Ashokan edicts mention a Mantri Parishad (council of ministers).** It is mentioned in the Arthashastra that the state cannot function without the assistance of ministers. Yet the Council had to report its opinion to the king immediately. **The primary role of the Council was advisory in nature. The king's decision was final in all respects. Hence statement 3 is not correct.**
- As per Megasthenes's Indica, there are numerous references to different committees in the district councils. Panyadhyaksha was responsible to look after trade and commerce and inspect weights and measures. **The collection of taxes was the responsibility of Sulkahyaksha and the registration of births and deaths was the work of Gopa.** The head of the urban administration was called Nagarika. He was assisted by two subordinate officials-Gopa and Sthanika. Other officials are also mentioned such as Bandhanagaradhyaksha (looked after the jail); Rakshi (i.e. the police; looked after the security of the people); Lohadhyaksha, Sauvarnika (officials who looked after goods that were manufactured in the centers). **Hence statement 2 is not correct.**

Q 6.D

- Kathakali, as a dance form popular today, is considered to be of comparatively recent origin. **However, it is an art which has evolved from many social and religious theatrical forms which existed in the southern region in ancient times. Chakiarkoothu, Koodiyattam, Krishnattam and Ramanattam are few of the ritual performing arts of Kerala** which have had a direct influence on Kathakali in its form and technique.
- In the temple sculptures in Kerala and the frescoes in the Mattancheri temple of approximately the 16th century, dance scenes depicting the square and rectangular basic positions so typical to Kathakali are seen.
 - **For body movements and choreographical patterns, Kathakali is also indebted to the early martial arts of Kerala. Hence statement 1 is correct.**
- **Kathakali is a blend of dance, music and acting and dramatizes stories**, which are mostly adapted from the Indian epics. It is a stylised art form, the four aspects of abhinaya - angika, aharya, vachika, satvika and the nritya, nritya and natya aspects are combined perfectly.
- The dancer expresses himself through codified hastamudras and facial expressions, closely following the verses (padams) that are sung. Kathakali derives its textual sanction from Balarama Bharatam and Hastalakshana Deepika.
- **Kathakali is a visual art where aharya, costume and make-up are suited to the characters, as per the tenets laid down in the Natya Shastra.** The characters are grouped under certain clearly defined types like the pacha, kathi, thadi, kari or minukku.
 - **The face of the artist is painted over to appear as though a mask is worn. The lips, the eyelashes and the eyebrows are made to look prominent. A mixture of rice paste and lime is applied to make the chutti on the face which highlights the facial make-up. Hence statement 2 is correct.**
- **Kathakali dance is chiefly interpretative. The characters in a Kathakali performance are broadly divided into satvika, rajasika and tamasika types.** Satvika characters are noble, heroic, generous and refined. In pacha, green colour dominates and kiritu (headgear) is worn by all. Krishna and Rama wear special crowns decorated with peacock feathers. The noble characters like Indra, Arjun and the Devas are some of the pacha characters. **Hence statement 3 is correct.**
- In no other dance style is the entire body used so completely as in Kathakali. The technical details cover every part of the body from facial muscles to fingers, eyes, hands and wrists. The facial muscles play an important part. The movement of the eyebrows, the eye-balls and the lower eye-lids as described in the Natya Shastra are not used to such an extent in any other dance style. The weight of the body is on the outer edges of the feet which are slightly bent and curved.
- **Kalasams are pure dance sequences where the actor is at great liberty to express himself and display his skills.** The leaps, quick turns, jumps and the rhythmic co-ordination make kalasams, a joy to watch.
- **A Kathakali performance begins with the kelikottu, calling the audience to attention followed by the todayam. It is a devotional number performed where one or two characters invoke the blessings of the gods.**
- Kelikottu is the formal announcement of the performance done in the evening when drums and cymbals are played for a while in the courtyard. A pure nritya piece known as the purappadu comes as a sequel to this.
- Then the musicians and drummers hold the stage entertaining the audience with an exhibition of their skills in melappada. Tiranokku is the debut on the stage of all characters other than the pacha or minukku. Thereafter, the play or the particular scene of the chosen play begins.

Q 7.B

- **Recent context:** Logged out of SWIFT (the messaging service to facilitate and confirm cross-border payments) most Russian banks are looking for alternative ways. India and several other countries too want a way out so that trade can continue. India and Russia are now exploring direct trade in rupee-ruble. And this is where Nostro and Vostro may come into play.
- **Nostro, along with Vostro, is a term used to describe a bank account. In fact, both terms are used to describe the same bank account.**
- These terms come into play when one bank has another bank's money on deposit. Typically, this is related to international trade and financial transactions. Nostro and vostro are derived from Latin words that mean "ours" and "yours," respectively.
- Example:
 - **Bank A will use the term Nostro account to refer to "our" account held by Bank B. Basically, Nostro is shorthand for "our money that is on deposit at your bank".** The Nostro account is the record maintained by the bank that has money deposited with another bank. They enable simplified trade settlements and foreign exchange transactions. Nostro accounts are denominated in foreign currencies.

- **Bank B, where bank A's money is deposited, will use the term Vostro account to refer to "your money that is on deposit at our bank".**
 - ✓ For example, if an Indian bank maintains an account with an overseas bank in the US in dollars, such an account, maintained in a foreign currency at a foreign center, would be called a Nostro account by the concerned Indian bank. The American bank concerned will refer to the same account as a Vostro Account.
 - ✓ Suppose a Russian bank opens a rupee account with a bank in India and puts Rs 10 lakh in it. Then India's liability becomes rouble 10.1 lakh as the exchange rate is 1.1. It will be a vostro account. Now, if an Indian businessman exports goods to a Russian buyer worth ₹10 lakh, then the Indian bank will debit the Russian bank's vostro account and the transaction will be squared up.
- **Hence option (b) is the correct answer.**

Q 8.B

- **Religion during Later Vedic period:**
 - Gods of the Early Vedic period like Indra and Agni lost their importance. Prajapathi (the creator), Vishnu (the protector), and Rudra (the destroyer) became prominent during the Later Vedic period. **Hence statement 1 is not correct.**
 - Sacrifices were still important and the rituals connected with them became more elaborate.
 - The importance of prayers declined and that of sacrifices increased. **Hence statement 2 is correct.**
 - The priesthood became a profession and a hereditary one. The formulae for sacrifices were invented and elaborated by the priestly class. Therefore, towards the end of this period, there was a strong reaction against priestly domination and against sacrifices and rituals.
 - The rise of Buddhism and Jainism was the direct result of these elaborate sacrifices. Also, the authors of the Upanishads, which is the essence of Hindu philosophy, turned away from the useless rituals and insisted on true knowledge (jnana) for peace and salvation.

Q 9.B

- The Junagarh prasasti also throws light on the diversified revenue demands of the state. This inscription describes that the treasury (kosa) of Rudradaman I overflowed with precious metals like gold (kanaka), silver (rajata), and gems by dint of his collection of lawfully levied taxes (yathavaprakta), like rent on land (bali), share on agricultural produce (bhaga), and tolls and customs (sulka). It also mentions that one of Chandragupta Maurya's governors, Pushyagupta, was responsible for building a dam on Sudarshana Lake near Girnar in Kathiawar. **Hence pair 1 is correctly matched.**
- The Rabatak inscription is an inscription written on a rock in the Bactrian language and the Greek script, which was found in 1993 at the site of Rabatak, near Surkh Kotal in Afghanistan. The inscription relates to the rule of the Kushan emperor Kanishka and gives remarkable clues on the genealogy of the Kushan dynasty. It dates to the 2nd century CE. **Hence pair 2 is correctly matched.**
- The Great Mauryan ruler Ashoka embraced Buddhism (as a part of shraman tradition) and the immense Buddhist missionary activities that followed during his rule paved the way for the development of Mauryan sculptural and architectural styles. The shraman tradition refers to several Indian religious movements parallel to but separate from the historical Vedic religion. Ashoka erected pillars and edicts all over the subcontinent and even in modern-day Afghanistan, Nepal, Bangladesh, and Pakistan to spread Buddha's word. One of which is the Rummindei Inscription mentions that the village of Lumbini (birthplace of the Buddha) was exempted from Bali and was to pay only one-eighth of the bhaga. **Hence pair 3 is correctly matched.**

Q 10.B

- The creator of the **modern Maratha nation was Shivaji Bhosale**, whose life, 1627-1680, bridges the gulf in Deccan history, between the extinction of the Ahmadnagar Sultanate and Aurangzeb's coming for the last time to the Deccan to wear out his life and empire there.
- Shivaji was born at Shivneri Fort in District Pune in the present-day state of Maharashtra. He was **formally crowned as Chhatrapati of Raigad on June 6, 1674**. Shivaji was the **son of Sahaji Bhosale**. **Hence, statement 1 is not correct.**
- The Battle of Pratapgad was a battle fought on 10 November 1659, at the fort of Pratapgad, near the town of Satara, Maharashtra, India, between the forces of the Marathas under Shivaji and the Adilshahi troops under the Adilshahi general Afzal Khan. **Hence statement 3 is not correct.**
- Mughal Emperor Aurangzeb was the sixth emperor of the Mughal Empire and ruled from July 1658 until his death in 1707. **Shivaji and Aurangzeb were contemporaries. Hence, statement 2 is correct.**

Q 11.C

- A distinct feature of the Vijayanagara state was the importance of the Brahmans as political and secular personnel rather than ritual leaders. Most of the durga dannaiks (in charge of forts) were Brahmans. Literary sources substantiate the theory that fortresses were significant during this period and were placed under the control of the Brahmans, especially of Telugu origins. **Hence, statement 1 is not correct.**
- The rulers of the early **Sangama dynasty were Saivas** who made additions to the Sri Virupaksha temple of Vijayanagara. The **Saluvas were basically Vaishnavas** who gave patronage to both the Siva and Vishnu temples. Krishnadeva Raya constructed the Krishnaswami temple (**Vaishnava shrine**) and also gave grants to Siva temples. The **Aravidu kings** also gave gifts to **Vaishnava temples**. **Hence, statement 2 is not correct.**
- The relationship between kings, sects, and temples can be explained in terms of four assertions:
 - Temples were basic for sustaining kingship.
 - Sectarian leaders were the connecting links between kings and temples.
 - Though the routine supervision of the temples was done by local sectarian groups, **the task of solving disputes concerning temples was in the hands of the king. Hence, statement 3 is correct.**
 - The intervention of the king in the above matter was administrative, not legislative

Q 12.A

- **Tyagaraja (1767 - 1847) is considered the most legendary Carnatic music composer**, who played a very significant role in the development of this music genre. **He saw music as a means to experience the love of god and as such, his sole intention was purely devotional while performing.**
 - **Tyagaraja created numerous devotional songs, most of which are devoted to the Hindu god Rama and are popular even today.** When Tyagaraja was merely 8 years old, he composed Nammo Nammo Raghavaya Anisham in raga Desikathodi.
- **Muthuswami Dikshitar (1775 - 1835) was an exponent of the South Indian Carnatic music genre. He created about 500 compositions in total, most of which are widely sung by renowned musicians in Carnatic music performances even today.**
 - A number of the compositions made by Muthuswami Dikshitar are in the Sanskrit language. They have been couched in the Krithi style i.e. a format in which the poetry is set to music. Throughout the course of his life history, Muttuswami Dikshitar toured a number of holy places in the country. And he's said to have composed Krithis on various deities and temples he visited.
 - Each and every one of the 500 compositions he's made is not only melodious but also full of depth. Though his Sanskrit compositions are themed on the temple gods and goddesses, all of them talk about the concept of advaitin i.e. the one with a form. These songs penned by Muthuswami Dikshitar talk much about the history of the temples and the customs and traditions observed within its precincts. Thus, they also serve as valuable sources of historical information.
- **Venkata Subrahmanya, affectionately known as Shyama Shastri (1762 - 1827) was born in a Tamil-speaking brahmin community known as auttara vadama.** His parents though scholarly, had no particular interest in music.
 - Although he did not compose as many kritis Shyama Shastri's compositions are well known. **It is said that he has composed about three hundred pieces in all. He composed in Telugu, Sanskrit, and Tamil and mostly on Goddess Devi.**
 - Shyama Shastri's compositions are usually in common ragas, except for a few in ragas such as Manji, Chintamani, Kalagada, and Karnataka Kapi.
- **Purandara Dasa (1484 - 1564) was a saint, musician, singer, social reformer, poet, and most importantly, a great devotee. Purandara Dasa followed Dvaita philosophy and was a disciple of Vyasa Thirtha.** He preached the divinity of the soul and worthlessness of pride of caste, among other things, through his simple and colloquial language, and understandable diction that could reach the masses.
 - **He was a virtuoso in Carnatic music. Such was his expertise in this field of music that he earned the epithet 'pitamaha' or the grandfather of this form of music.**
- **Thus the birth of the Musical Trinity - Tyagaraja, Muthuswami Dikshitar and Syama Sastri - at Tiruvarur between the years 1750 to 1850 A.D. ushered in an era of dynamic development in Carnatic music.**
 - **The Trinity were not only contemporaries among themselves but, also contemporaries of great composers of Western Music, as Beethoven, Mozart, Wagner and Haydn.**
- **Hence option (a) is the correct answer.**

Q 13.C

- The most significant source of the Mauryan period is undoubtedly the inscriptions of Ashoka. Ashoka's inscriptions mark the beginning of Indian epigraphy. What sets Ashoka's edicts apart is that they are issued in the first person, thereby revealing the voice and ideas of the king himself.
- The **edicts are written in the Prakrit language and Brahmi script, and occasionally in the Kharoshti script (in the northwestern parts of the subcontinent). There are a few inscriptions in Greek and Aramaic as well.** A bilingual Greek-Aramaic inscription was found at Shar-iKuna near Kandahar in southeast Afghanistan and one in Taxila. **Hence statement 1 is correct.**
- **Ashoka himself had designated these edicts as Dhammalipi (Edicts of Piety). The Maski edict in the Prakrit language, carved in Brahmi script and dated 256 BC, remains a dharma shasana, an edict exhorting people to follow the tenets of Buddhism. The Maski edict clearly told the world that it was Ashoka who had got the inscriptions carved under the name 'Devanampriya'. The inscription has a mention of 'Devanampriya Asoka.'** Apart from associating the title 'Devanampriya' with Ashoka, the inscription suggests the spread of Mauryan rule up to the Krishna valley of north-eastern Karnataka. **Hence statement 2 is correct.**

Q 14.C

- Science and Mathematics were highly developed during the ancient period in India. Ancient Indians contributed immensely to the knowledge in Mathematics as well as various branches of Science.
- **Bhaskaracharya** was the leading mathematician of the 12th Century. He was born at Bijapur, Karnataka. **He is famous for his book Siddhanta Shiromani. It is divided into four sections:**
 - **Lilavati (Arithmetic)**
 - **Beejaganit (Algebra)**
 - **Goladhyaya (Sphere), and**
 - **Grahaganit (mathematics of planets).**
- **Bhaskara introduced Chakrawat Method or the Cyclic Method to solve algebraic equations.** This method was rediscovered six centuries later by European mathematicians, who called it inverse cycle. In the nineteenth century, an English man, James Taylor, translated Lilavati and made this great work known to the world.
- **Baudhayana** was the first one ever to arrive at several concepts in Mathematics, which were later rediscovered by the western world.
 - The value of pi was first calculated by him. **What is known as Pythagoras theorem today is already found in Baudhayana's Sulva Sutra, which was written several years before the age of Pythagoras.**
- In the 7th century, **Brahmagupta** took mathematics to heights far beyond others. In his methods of multiplication, he used place value in almost the same way as it is used today. He introduced negative numbers and operations on zero into mathematics. **He wrote Brāhmasphuṭasiddhānta** through which the Arabs came to know our mathematical system.
- There is an elaborate description of mathematics in Jain literature (500 B.C -100 B.C). Jain gurus knew how to solve quadratic equations. They have also described fractions, algebraic equations, series, set theory, logarithms and exponents in a very interesting manner.
 - **Jain Guru Mahaviracharya wrote Ganit Sara Sangraha in 850A.D.,** which is the first textbook on arithmetic in present-day form. The current method of solving the Least Common Multiple (LCM) of given numbers was also described by him. Thus, long before John Napier introduced it to the world, it was already known to Indians.
- **Hence option (c) is the correct answer.**

Q 15.D

- After the decline of the Mauryan Empire, northwest India was constantly under attack by various invaders from Central and West Asia. **The Sakas (also written Sakas), alternatively known as Indo-Scythians, invaded northwest India from the first century BC onwards.**
- There were five branches of the Shakas with their seats of power in different parts of India and Afghanistan.
 - One branch of the Shakas settled in Afghanistan. Prominent rulers of this branch were Vonones and Spalirises.
 - The second branch settled in Punjab with Taxila as its capital. Maues was a prominent ruler.
 - The third branch settled in Mathura, where they ruled for about two centuries. Azilises was a prominent ruler.
 - **The fourth branch established its hold over western India, where they continued to rule till the 4th century CE.**

- The fifth branch of the Shakas established its power in the upper Deccan.
- The **fourth branch of Sakas ruled for the maximum period owing to a flourishing economy based on the sea-borne trade in Gujarat and also issued a large number of silver coins.** One of the famous Shaka rulers was **Rudradaman 1** (CE 130-150). He ruled over Sindh, Kutch, and Gujarat and also recovered from the Satavahanas, Kokan, the Narmada valley, Malwa and Kathiawar. He is famous in history because of the repairs he undertook to improve **Sudarsana lake in the semi-arid zone of Kathiawar.** He was a **great lover of Sanskrit and issued the first-ever long inscription in chaste Sanskrit.**
- The Shakas did not meet effective resistance from the rulers and the masses of India. **The king of Ujjain (around 58 BCE), effectively fought and succeeded in throwing the Shakas out. He called himself Vikramaditya and an era called Vikram-Samvat is reckoned from the event of his victory over the Shakas in 58 BCE. Hence statement 1 is not correct.**
- From this time onwards, Vikramaditya became a coveted title, and whoever achieved anything great adopted this title, as the Roman emperors adopted the title Caesar in order to emphasize they are a great power.
- **Kanishka, who was a Kushan ruler, extended his whole-hearted patronage to Buddhism. He also convened the fourth Buddhist council to discuss matters relating to Buddhist theology and doctrine. It was held at the Kundalavana monastery near Srinagar (Kashmir) under the presidency of Vasumitra. It was in this council that Buddhism was split into two schools – the Hinayana and the Mahayana. Hence statement 2 is not correct.**

Q 16.A

- **Recent context:** Marine researchers record a new marine flatworm in Rushikonda beach of Visakhapatnam, the first such documented record of the species as well as the order from the east coast of India.
- **The vibrant-colored benthic organism was a species of polyclad flatworm Pseudoceros galatheensis, recorded for the first in Andhra Pradesh and East Coast mainland India. Hence option (a) is the correct answer.**
- Marine flatworms, also called polyclads, are usually seen in nearshore areas in tropical and subtropical areas.
- They are predatory and mainly feed on marine organisms like sponges, ascidians, crabs, and other smaller organisms. Their role is vital as predators in coral reefs and other shallow water marine ecosystems.
- Their brilliant color warns other predators that they are toxic and should not be consumed.
- This species was first identified in the Andamans in 2017. These organisms are usually seen in rocky as well as intertidal areas.
- They have two folds on the anterior side called pseudo tentacles with around 12 eye spots on each fold that are used for sensing light. As thin as paper, they can surprisingly prey upon organisms like crabs by hunting them.
- Though there are many recorded species of polyclad flatworms around the world, very less is known about them in the east coast of India, where the records were not described until recently.

Q 17.D

- **The cultures were based on the use of copper and stone implements called Chalcolithic.** Technologically, the Chalcolithic stage is applied to the pre-Harappan phase. However, in various parts of India, the Chalcolithic cultures followed the Bronze Age Harappa culture. The Chalcolithic people mostly used stone and copper objects, but they also occasionally used low-grade bronze and even iron.
- The Chalcolithic people domesticated cattle, sheep, and goats, which were tethered in the courtyard. **In all probability, the domesticated animals were slaughtered for food and not used for milk and dairy products.** The tribal people, such as the Gonds of Bastar, believe that milk is meant only to feed young animals, and, therefore, they do not milk their cattle. Consequently, the Chalcolithic people were not able to make full use of the animals. **Hence, statement 1 is not correct.**
- Ganeshwar is a Chalcolithic site in Rajasthan whose deposits are ascribed to 2800–2200 BC, they, by and large, predate the mature Harappan culture. **Ganeshwar principally supplied copper objects to Harappa and did not receive much from it.** The Ganeshwar people partly lived on agriculture and largely on hunting. Although their principal craft was the manufacture of copper objects, **they were unable to urbanize.** With its microliths and other stone tools, much of the Ganeshwar culture can be considered a pre-Harappan Chalcolithic culture that contributed to the making of the mature Harappan culture. **Hence, statement 2 is not correct.**

Q 18.A

- India's agricultural Census has been conducted from 1970-71 in collaboration with the States and Union Territories as part of a **program of the World Census of Agriculture of FAO. Hence statement 2 is correct.**
 - **It is conducted every 5 years. Hence statement 1 is correct.**
- It is the main source of information on a variety of indicators, including the nature of operational land holdings, their size, distribution by class, land-use statistics, tenancy, and Cropping patterns. **Hence statement 3 is correct.**
 - Census provides the basis for the development of a comprehensive integrated national system of agricultural statistics and has links with various components of the national statistical system.
- **During the 11th census, for the first time, data collection will be conducted on smartphones and tablets.**
 - This will help in faster and more accurate enumeration.
 - Most states have digitized land records and surveys, which will further accelerate the collection of agricultural census data.
- **According to the 10th agricultural census (2015-16):**
 - Small and marginal farmers (less than two hectares of land) accounted for 86.2% of all farmers but owned just 47.3% of the crop area.
 - Land holdings continued to be inequitably distributed.
- **World Census of Agriculture (WCA):**
 - The first WCA was carried out in 1930 by the International Institute of Agriculture. Beginning in 1950, the Food and Agriculture Organization (FAO) of the United Nations has been preparing and advocating the decennial WCA.
 - The concept of agricultural operational holdings adopted in India differs to some extent from that of FAO, as it does not include those holdings which are not operating any agricultural land and are engaged exclusively in livestock, poultry, and fishing, etc.

Q 19.A

- The **Maurya period constitutes a landmark in the system of taxation in ancient India.** Kautilya names many taxes which were collected from peasants, artisans, and traders. This required a strong and efficient machinery for assessment, collection, and storage. **The Mauryas attached greater importance to assessment than to storage and deposit. Hence statement 1 is correct.**
- The **samaharta was the highest officer in charge of assessment and collection, and the sannidhata was the chief custodian of the state treasury and storehouse.** The assessor- cum-collector was far more important than the chief treasurer. The damage inflicted on the state by the first was thought to be more serious than any inflicted by the second. **Hence statement 2 is not correct.**
- The epigraphic evidence for the existence of rural storehouses shows that taxes were also collected in kind. These granaries were probably also meant to help local people in times of famine, drought, etc.
- From **Ashoka's inscription at Lumbini, it was inferred that land revenue was of two kinds — Bali and Bhaga.** The assessment of the tax varied from region to region, from being on 1/6th to a quarter of the produce of the land. 1/4th of the produce was paid in tax by the peasants. They also paid a tribute. Land tax (bhaga) was the main source of revenue. It was levied at 1/6th of the produce. **The Lumbini edict of Ashoka says that during his visit to the birthplace of the Buddha, he exempted the village from the payment of bali and reduced the payment of Bhaga to 1/8th.**

Q 20.C

- The land in between the hills of Venkatam and Kanyakumari is called Tamilaham or Tamilakam. It includes the whole of the modern Tamil Nadu and Kerala. With forested hills, undulated terrains, pastures, arid zones, wetlands, and long sea coast, the region represented a combination of diverse eco-zones. **The three principal chiefdoms – the Cheras, Cholas, and Pandyas – had their strongholds both in the interior as well as on the sea coast.**
- The **Cheras** had Karur in the interior and **Mujiris, the well-known ancient port on the west coast.** The **Cholas** had Uraijur in the interior and **Puhar on the Coromandel coast as their strongholds.** Similarly, the **Pandyas** had Madurai and **Korkai as their interior headquarters and port respectively.** These were the most important political centres of the period in the region. **Hence pairs 1, 2 and 3 are correctly matched.**



Q 21.B

- **Paitkar paintings** : One of the most ancient schools of painting in the country, Paitkar paintings are a creative expression of the state's folk art. Popularly known as the scroll paintings of the east, **Paitkar paintings are mostly based on themes drawn from Hindu epics like Mahabharata and Ramayana.**
 - These traditional paintings tell the stories of the miracles performed by various gods and goddesses like Shiva and Durga. Paitkar painters usually make use of only primary colours like red, yellow and blue with palm leaves acting as a base.
 - The brushes used to paint are made from the hair of squirrels and goats. Most of the painted space in Paitkar paintings is occupied by human characters with elongated eyes, a prime characteristic of the Indian painting style.
 - **Located in the eastern part of Jharkhand, Amadubi Village is home to families of talented Paitkar artists. The village is also known as the village of Paitkars and the art form is said to have originated here. The Santhal tribe of the state believes that Paitkar paintings can send wandering souls of dead people to heaven. Hence pair 1 is correctly matched.**
- **Phad is an indigenous painting style of the state of Rajasthan, India, that depicts the narratives of the folk deities of the state. This style of painting is the blend of Rajput and Mughal styles of painting.** Originated hundreds of years ago, this fabled heritage of Bhilwara is done on a long piece of canvas called Phad. Mostly, the stories of the folk deity Pabuji and the Gurjar Warrior, Devnarayanji are depicted on Phads.
 - **Phad paintings are very detailed. Every inch of the canvas is used. Since the ancient stories are depicted in the paintings, Phads have a lot of human figures in them. The size and color of which depends upon the role and position they had. These paintings are done on clothes.**
 - A thick fine paste of wheat/rice flour, prepared by boiling the mixture in water, is applied to the cloth which is then dried in sunlight. The cloth is then rubbed with Mohra (a stone device) to bring out the smoothness and shine it. Now the cloth is ready to be painted. The colors used in Phad are all-natural, obtained from various plants and vegetable extracts. Earthen colors are used to bring the acrylic effect. **Hence pair 2 is correctly matched.**
- **Orissa Pattachitra** is an ancient painting art form of rural Orissa that has been in existence for many centuries. The word pattachitra has evolved from the Sanskrit words 'Patta' and 'Chitra'. In the Sanskrit language, the word Patta means "Canvas" or "A Piece of Cloth", and Chitra means picture.
 - **Orissa Pattachitra (Traditional Indian Painting) is a painting especially inspired by Lord Jagannath and temple tradition, the Puri district.**
 - Themes for these paintings range from the events of Krishna's life and the Avatars of Vishnu to the epic tales of Ramayana and Mahabharata.
 - The method of production for Pattachitra is very long and the entire process is completed in many steps, which include:

- ✓ Preparation of Patta
- ✓ Preparation of Pigments
- ✓ Motif Selection and Layout
- ✓ Painting
- **Orissa Pattachitra paintings hold a Geographical Indication tag. Hence pair 3 is not correctly matched.**

Q 22.A

- The Mauryan state took various measures to deal with emergency situations like drought and famine.
- The **Mahasthan inscription** of the Mauryan period records an order issued by a ruler to the **official called mahamatra** stationed in the area.
- This order mentions different types of famine relief measures undertaken by state officials. This includes 'gandakas' i.e. **loan advances in coins** and distribution of paddy from the granary.
- **Hence option (a) is the correct answer.**

Q 23.B

- The **six-member commission headed by Jayant Kumar Banthia, a former chief secretary, was set up by the Uddhav Thackeray-led Maha Vikas Aghadi government after the 27 percent reservation in local self-government bodies was struck down by the Supreme Court.**
- On July 20, 2022, The **Supreme Court of India accepted the report of the Jayant Kumar Banthia Commission.** Court also gave direction to Maharashtra State Election Commission to hold elections in accordance with the report.
- **Commission has recommended providing 27% reservation in local bodies.** This recommendation has been accepted by the supreme court. Thus, OBCs will get 27% political reservation in local self-government elections to be held in state.
- **G. Rohini commission was set up on October 2, 2017, under Article 340 of the Constitution. It was tasked with sub-categorization of the OBCs and equitable distribution of benefits reserved for them, especially with reference to OBCs included in the Central list.**
- **Hence option (b) is the correct answer.**

Q 24.A

- The **Hoysala dynasty ruled South Karnataka between the 11th and 14th centuries.** Their architecture is **considered an intermediate between the Indo-Aryan style and the Dravidian style of temples.** Hence **statement 1 is correct.**
- One of the **important characteristics of these temples is their star-shaped base, over which the main structure is raised on a platform,** the temple is also covered with intricate sculptural carvings. Hence **statement 2 is correct.**
- Another characteristic feature of these temples is that **they are made of soapstones instead of sandstones,** which enabled the intricate craftsmanship- **Hence statement 3 is not correct.**
- **The Hoysala Temples of Belur, Halebid, and Somnathapura in Karnataka are now on the final list of India's nominations for the World Heritage List for the year 2022-2023.**

Q 25.B

- **Social Life of Rig Vedic Society**
 - **The Rig Vedic society was patriarchal.** The basic unit of society was the family or kula. The head of the family was known as grahapathi.
 - **Monogamy was generally practiced while polygamy was prevalent among the royal and noble families.**
 - The wife took care of the household and participated in all the major ceremonies. **Women were given equal opportunities as men for their spiritual and intellectual development. Hence statement 2 is not correct.**
 - There were women poets like Apala, Viswavara, Ghosa and Lopamudra during the Rig Vedic period. Women could even attend the popular assemblies. **There was no child marriage and the practice of Sati was absent. Hence statement 3 is correct.**
 - Both men and women wore upper and lower garments made of cotton and wool. A variety of ornaments were used by both men and women. Wheat and barley, milk and its products like curd and ghee, and vegetables and fruits were the chief articles of food. The eating of cow's meat was prohibited since it was a sacred animal. Chariot racing, horse racing, dicing, music, and dance were the favourite

pastimes. **The social divisions were not rigid during the Rig Vedic period as it was in the later Vedic period.**

- **Religion**
 - The Rig Vedic Aryans worshiped natural forces like earth, fire, wind, rain and thunder. They personified these natural forces into many gods and worshipped them.
 - The important Rig Vedic gods were Prithvi (Earth), Agni (Fire), Vayu (Wind), Varuna (Rain), and Indra (Thunder). **Indra was the most popular among them during the early Vedic period. Next in importance to Indra was Agni who was regarded as an intermediary between the gods and people.** Varuna was supposed to be the upholder of the natural order.
 - There were also female gods like Aditi and Ushas.
 - **There were no temples and no idol worship during the early Vedic period. Hence statement 1 is correct.**
 - Prayers were offered to the gods in the expectation of rewards. Ghee, milk and grain were given as offerings. Elaborate rituals were followed during the worship.

Q 26.B

- **Kanishka was the most powerful ruler of the Kushana Empire. The capital of his empire was Purushpura (Peshawar).** Under his rule, Kushana Empire extended from Uzbekistan, and Tajikistan to Mathura and Kashmir. Kanishka was the successor of Vima Kadphises, as demonstrated by an impressive genealogy of the Kushan kings, known as the Rabatak inscription.
- **It was during Kanishka's reign that Buddhism was divided into Mahayana and Hinayana. He was the founder of the Saka Era of A.D. 78.** He had invaded Patliputra and had taken the Buddhist monk Asvaghosa to Purushpura. Kanishka was a patron of Buddhism and he called the 4th Buddhist council in the Kundalvana of Kashmir in 78 AD. The council was chaired by Vasumitra and during this council the collection of Buddhist texts took place and the commentaries were engraved on copper sheets.
- **Kanishka is also known for erecting a huge stupa and a monastery in Pakistan in today's Shaji-ki-Dheri, located in the outskirts of Pakistan.**
- **Hence option (b) is the correct answer.**

Q 27.A

- Sangam literature can be dated from its earliest to the first two centuries CE. **The poets were men and women drawn from all classes in society and they were richly rewarded for their compositions.** The composition of the texts took place during the three Sangams. **The word Sangam refers to a confluence or academy. Hence statement 2 is not correct.**
- **The three Sangams were patronized by Pandyan rulers and were called the Talai Sangam, Idai Sangam and Kadai Sangam respectively meaning the beginning, middle and end Sangams. Hence statement 1 is correct.**
- Sangam literature is a rich source because it describes a range of activities that were related to agriculture. Many secondary production activities, like the making of sugar from sugar cane, find rich descriptions in Sangam literature. Primary agrarian activities like the cultivation of ragi and sugarcane, the harvesting of grain and smaller activities like the drying of grain are also described in Sangam literature.
- The compositions of the first two Sangams are lost. **All the texts that make up the Ettutogai come from the Kadai or end Sangam.**
- The Sangam literature consists of **Tolkappiyam, Ettutogai, Pattuppattu, Pathinenkilkanakku and the two epics Silappathigaram and Manimegalai.** Among these the Tolkappiyam authored by Tolkappiyar was the earliest work, provides the information on Social, Economic and political conditions of the Sangam Age along with the Tamil grammar. **Ettutogai were the eight Anthologies consisting of eight works.** Both Ettutogai and Pattuppattu were divided into two main groups – Aham (love) and Puram (valour).

Q 28.C

- Diwan-i-Insha: During the Delhi Sultanate, Diwan-i-Insha was in charge of the royal correspondence, and ranked third in the administrative structure. His assistants used to make all correspondence, even confidential matters. Diwan-i-risalat or the minister of foreign affairs was in charge of dealing with diplomatic correspondences and the ambassadors.
- Diwan-i-Wizarat was the finance department headed by the wazir (Prime-minister). **Hence pair 1 is not correctly matched.**
- **Khalisa refers to land owned and administered directly by the state, those of which the revenue remains the property of the government, not being made over as Jagir or Inam to any other party, during the Delhi Sultanate and Mughals period. Hence pair 2 is correctly matched.**

- **Mamluk/Mameluk was a slave soldier**, a member of one of the armies of slaves established during the Abbasid era that later won political control of several Muslim states. The Mamluk dynasty was founded in Northern India by Qutb ud-Din Aibak, a Turkic Mamluk slave-general of the Ghurid Empire from Central Asia. **Hence pair 3 is correctly matched.**

Q 29.B

- The reign of Chandragupta II saw the high watermark of the Gupta Empire. Chandragupta II adopted the title of Vikramaditya. **An important incident that took place during his reign was the visit of Fa-Hien**, a Chinese pilgrim, who came to India in search of Buddhist texts. Numerous scholars adorned the court of Chandragupta II at Ujjain.
- It is believed that **Chandragupta II was deeply interested in the art and culture and Navratnas that adorned his court**. A few of them are as follows:
 - **Amarasimha**: He was a Sanskrit lexicographer and a poet.
 - **Dhanvantri**: He was believed to be a great Physician who worked in the court of Chandragupta II.
 - **Harisena**: He is known to have composed the Allahabad pillar Inscription. In his old age, Harisena was in the court of Chandragupta II and described him as a Nobel king.
 - **Kalidasa**: Kalidasa is known to be an immortal poet and playwright of India.
 - **Kahapanka**: He was known to be an astrologer working in the court of Chandragupta II.
 - **Sanku**: He was in the field of architecture and contributed a lot to the beautiful structures during the Chandragupta II reign.
 - **Varahamihira**: he wrote several books, and three of the most important were: Brihat Samhita, Panchasiddhantika, and Brihat Jataka.
- **Hiuen Tsang or Xuanzang was a Chinese Buddhist monk who traveled overland from China to India during the reign of King Harsha Vardhan to obtain Buddhist scriptures.**
- **Hence option (b) is the correct answer.**

Q 30.A

- The **Sangam Age is significant and unique for the social, economic, religious, and cultural life**. There was an all-around development during this period. The Sangam literature as well as the archaeological findings reveal these developments.
- Society during the Sangam period was broadly divided into several groups. At the beginning of the Sangam Age, society was not organized on the basis of the Vedic caste system, namely Brahmanas, Kshatriyas, Vaisyas, and Sudras. However, the earliest of the Sangam literature, Tolkappiyam refers to the four divisions prevalent in the Sangam society namely, Anthanar, Arasar, Vaisiyar, and Vellalar. It may be said that this classification roughly corresponds to the Vedic social division.
- The **ruling classes were called Arasar, and its members had marriage relations with the Vellalas, who formed the fourth caste**. Agricultural operations were generally carried on by members of the lowest class (Kadaiyiar), whose status appears to have differed little from that of the slave.
- Several out-castes and forest tribes suffered from extreme poverty and lived from hand to mouth. We notice sharp social inequalities in the age of the Sangam. **The rich lived in houses of brick and mortar and the poor in huts and humbler structures. In the cities, the rich merchants lived in the upper storey of their houses. Hence statement 1 is correct.**
- In the Sangam Age, women were treated with special consideration. Natural feminine qualities such as Achcham, Madam, and Nanam were insisted on in the Sangam literature. Their most important virtue was chastity. The heroine of Silappathigaram, Kannagi had been hailed for her chastity and worshipped by the people. The women were given the freedom to choose their life partners during the Sangam period.
- Women were not permitted to remarry and inherit property. **Sati or the custom of self-immolation at the death of one's husband was not generally prevalent during this period. However, some women from the royal family indulged in the practice of Sati. Hence statement 2 is not correct.**

Q 31.D

- In the Kannada region, the Bhakti movement begun by Basavanna (1105-68) in the 12th century for a time threatened the caste hierarchy and stretched the fabric of local society.
- **Basavanna initially was a Jaina and the minister of Kalachuri King Bijjala II (a vassal of Chalukyas)**. He used his considerable powers to initiate a programme of social reform. **Hence statement 1 is correct.**
- His **followers were known as Virashaivas (heroes of Shiva) or lingayats (wearers of linga)**. They worship **Shiva in his manifestation as linga**, and **men usually wear a small linga in a silver case on a loop strung over the left shoulder. Hence statement 2 is correct.**

- **Bhakti movement in this region produced a rich vein of literature that came to be known as Vachana Sahitya, composed by Basava himself as well as his disciples (Akkamahadevi, Allama Prabhu, Devara Dasimayya, and others). Consisting of pithy aphorisms, these Vachanas conveyed specific astute observations on spiritual and social matters in unambiguous terms. Hence statement 3 is correct.**

Q 32.B

- Hambantota port: Sri Lanka's Defence Ministry recently denied reports that a Chinese research vessel involved in space and satellite tracking would enter the Hambantota port in August this year. The Hambantota International Port is a deep water port in Hambantota, Sri Lanka. It is Sri Lanka's second largest port, after the Port of Colombo. **Hence pair 1 is correctly matched.**



- India and Bangladesh are preparing for comprehensive trial runs for trans-shipment of cargo to the northeastern states using Chattogram and Mongla ports as part of efforts to further boost connectivity between the two sides. Chittagong, officially called Chattogram, city that is the chief Indian Ocean port of Bangladesh. In the year 2010, India and Bangladesh signed a Memorandum of Understanding (MoU) to allow for the use of Chittagong and Mongla Ports in Bangladesh for the movement of goods to and from India. **Hence pair 2 is correctly matched and pair 3 is not correctly matched.**



Q 33.D

- **Lalitavistara is a biography of the legendary life of the Gautama Buddha, written in a combination of Sanskrit and vernacular.**
- The book originated from the Pāli scriptures of early Buddhism and was later expanded and incorporated into the Mahāyāna tradition.
- Like the Mahavastu ("Great Story"), the subject matter of which is the same, the Lalitavistara contains late material but also preserves some very ancient passages. It shares with the Hindu Puranas similarities of style as well as the concept of a divine being's earthly activities as "sport," or "play."
- In characteristic Mahayana fashion, an introductory chapter describes the Buddha, deep in meditation and surrounded by a divine effulgence, about to reveal the contents of the text to an assemblage of monks and bodhisattvas.
- In the ensuing narrative, it is especially with regard to the Buddha's conception and birth that this work adds to the miraculous and mythological elements of earlier accounts. The Lalitavistara has inspired a considerable amount of Buddhist art.
- **Hence option (d) is the correct answer.**

Q 34.C

- **Vijayanagara architecture:** Vijayanagara architecture of 1336–1565 CE was a famous building idiom that developed during the rule of the imperial Hindu Vijayanagara Empire. The empire ruled South India, from their regal capital at Vijayanagara, on the banks of the Tungabhadra River in modern Karnataka, India.
 - Architecture attained a certain fullness and freedom of expression during the Vijaynagar rule. Though often characterized as Dravida style, it had its own distinct features. This new style of architecture **called as Provida style. Hence, statement 1 is correct.**
 - **Features of temples:**
 - ✓ Large temple complex
 - ✓ Temples were elaborate structures and process of horizontal elaboration continued.
 - ✓ Ornamentation became rich and heavy.
 - ✓ Huge compound wall.
 - ✓ A new structure known as **Amman Shrine** appeared. Here spouse of the chief deity was kept. eg: **Amman shrine of Hazara temple. Hence, statement 2 is correct.**
 - ✓ **Mandapa:** Mandapa or open pavilion with a raised platform meant for sitting deities. The new structure was **Kalyan Mandapa** where the union of God and spouse was done on special occasions. **Thousand Pillar mandap** became popular. It was a huge hall having numerous rows of pillars.
 - ✓ **Pillars:** Unique pillars which are highly carved and characterized by various kinds of designs. eg: **Pillar hall at Lepakshi temple.**
 - ✓ **Gopurams:** Most Gopurams were added by the most famous king of the Vijaynagara empire, Krishna Deo Raya. The Gopurams were in several storied pyramidal structures. Larger and taller Gopurams are known as **Raya Gopuram. Hence, statement 3 is correct.**
 - The climax of temple architecture at Vijayanagara occurred under the Tuluva rulers. The architectural tradition was accompanied by a vibrant sculptural tradition that used many mythological figures and narratives. The shrines on Hemakuta hill, **Virupaksha temple, and the Hazara Rama temple** are examples of Vijayanagara temple architecture.
 - Krishnadevaraya built some fine stone temples and added impressive gopurams or gateways to many important South Indian temples. He also founded a suburban township near Vijayanagara called Nagalapuram after his mother. Some of the most detailed descriptions of Vijayanagar come from his period. **Tirupati's famous temple** developed greatly during his period as the deity there was his titular deity.

Q 35.A

- When Pallavan king Simhavishnu (500-580 CE) was busy expanding his territories, it was almost about the same time the Chalukyas of Badami started ruling in North Karnataka with Badami (Bijapur District) as their capital. **The founder, Pulakesin I (543-66 CE)** converted the hill near Badami into a strong fortress and launched his expansionist activities. **Hence, statement 1 is correct.**
 - The territory of the Kadambas of Banavasi towards the South and that of the Mauryas of Konkan on the West were soon conquered and annexed to their growing territory by Kirtivarman I (566-597 CE).
- **Pulakesin II** came head-on against his greatest northern adversary, **Harsha of Kanauj**, who was planning to attack the Deccan and won a decisive victory on the banks of the Narmada. **Ravikirti's Aihole**

inscription speaks in detail the victory of Pulakesin II over Harsha. After the victory Pulakesin II assumed the title of **Parameshvara (Supreme Lord)**. Hence, **statement 2 is correct.**

- The Chalukyan rule in Badami was replaced by the Rashtrakutas in about 750 CE when Rashtrakuta feudatory Dantidurga gave a final blow and defeated the **Chalukyan king Kirtivarman II**. Hence, **statement 3 is not correct.**

Q 36.A

- **BODHISATTVA IN MAHAYANA BUDDHISM**

- Bodhisattva is the one who has generated Bodhicitta (a spontaneous wish and a compassionate mind) to attain Buddhahood for the benefit of all sentient beings. It believes in Universal liberation and is a concept under Mahayana Buddhism.
- Buddha in his previous births was Bodhisattva as contemplated in Jataka tales and the stories depict the various attempts of the bodhisattva to embrace qualities like self-sacrifice and morality.
- Under Theravada Buddhism, a person whose aim is to become fully enlightened is still subject to birth, illness, death, sorrow, defilement, and delusion. On the path to become a Buddha, a bodhisattva proceeds through ten grounds or bhumis namely Great Joy, Stainless, luminous, radiant, very difficult to train, obviously transcendent, gone afar, immovable, Good discriminating wisdom, and Cloud of Dharma. By passing these 10 bhumis, he becomes an enlightened one.

- **Prominent Bodhisattvas under Buddhism include:**

- **Avalokitesvara:** One of the three protective deities around Buddha. He is described as holding Lotus flower and is also known as Padmapani. The painting can be found at Ajanta Caves. He is the most acknowledged among all the Bodhisattvas. The bodhisattva of compassion, the listener of the world's cries who uses skillful means to come to their aid. He appears unofficially in Theravada Buddhism in Cambodia under the name Lokeshvara. He is depicted as a female also and is said to incarnate in his holiness the Dalai Lama.
- **Vajrapani:** One of three protective deities around Buddha and is also depicted in Ajanta Caves. Vajrapani is contemplated to manifest all the powers of Buddha as well as the power of all five tathagatas namely Vairocana, Akshobhya, Amitabha, Ratnasambhava and Amoghasiddhi.
- **Manjusri:** One of three protective deities around Buddha is also depicted in Ajanta Caves. **He is associated with the Wisdom of Buddha and is a male Bodhisattva with a wielding sword in his hand. Hence pair 1 is correctly matched.**
- **Samantabhadra:** associated with practice and meditation. Together with the Buddha and Manjusri, he forms the Shakyamuni trinity in Buddhism. **Hence pair 2 is not correctly matched.**
- **Ksitigarbha:** He is depicted a Buddhist monk and took vow not to achieve Buddhahood till the hell is completely emptied.
- **Maitreya:** A future Buddha who will appear on Earth in the future, achieve complete enlightenment, and teach the pure dharma. Laughing Buddha is said to be an incarnation of Maitreya.
- **Akasagarbha:** associated with element of space. **Hence pair 3 is not correctly matched.**
- **Tara:** associated only with Vajrayana Buddhism and represents the virtues of success in work and achievements.
- **Vasudhara:** associated with wealth, prosperity, and abundance. Popular in Nepal. **Hence pair 4 is correctly matched.**
- **Skanda:** Guardian of viharas and the Buddhist teachings.
- **Sitatapatra:** She is contemplated as a protector against supernatural danger and is worshipped in both Mahayana and Vajrayana traditions.

Q 37.C

- The tradition of constructing pillars is very old in India and it may be observed that the erection of pillars was prevalent in the Achaemenian empire as well. But the Mauryan pillars are different from the Achaemenian pillars. The Mauryan pillars are rock-cut pillars thus displaying the carver's skills, whereas the Achaemenian pillars are constructed in pieces by a mason.
- Stone pillars were erected by Ashoka, which have been found in the north Indian part of the Mauryan Empire with inscriptions engraved on them. The top portion of the pillar was carved with capital figures like the bull, the lion, the elephant, etc. All the capital figures are vigorous and carved standing on a square or circular abacus. Abacuses are decorated with stylized lotuses.
 - Some of the existing pillars with capital figures were found at Basarah-Bakhira, Lauriya Nandangarh and Rampurva in Bihar, Sankisa and Sarnath in Uttar Pradesh.
- **The Mauryan pillar capital found at Sarnath popularly known as the Lion Capital is the finest example of Mauryan sculptural tradition. It is also our national emblem. It is carved with**

considerable care—voluminous roaring lion figures firmly standing on a circular abacus which is carved with the figures of a horse, a bull, a lion and an elephant in vigorous movement, executed with precision, showing considerable mastery in the sculptural techniques.

- This pillar capital symbolising Dhammachakrapravartana (the first sermon by the Buddha) has become a standard symbol of this great historical event in the life of the Buddha.
- Hence option (c) is the correct answer.

Q 38.C

- **Developments of Paintings during the Mughal Period:**
 - **Humayun:** He brought back with him the master artists when he regained power in India. He invited two Persian artists — **Mir Sayyid Ali and Abd us Samad** to establish a studio in his court and carry out royal paintings. It is important to note here that both the artists were famous and respected, particularly for their skills in the art of portraiture. **Hence option (c) is the correct answer.**
 - **He founded the Nigaa Khana** (painting workshop), which was also a part of his library. However, it is known that he started the project of illustration of **Hamza Nama** that was continued by his son and successor Akbar.
 - **Babur:** He had a keen eye for portraiture and this is also recorded in his memoirs. Among the artists, who find mentioned in Babur's memoirs is Bihzad. **Bihzad's work** was dainty but he did not draw the faces well; he used to greatly lengthen the double chin (ghab-ghab); and drew bearded faces admirably. Bihzad was a master artist from the Persian school of painting, Herat (now in present-day Afghanistan), and was known for his sophisticated compositions and color tints. Also, **Shah Muzaffar** finds mentioned as a painter, who Babur thought excelled in the representation of hairstyle.
 - **Akbar:** He commissioned the translation and illustration of revered Sanskrit texts into Persian. The **Persian translation and illustrated version of the Hindu epic Mahabharata** during this period **came to be known as Razm Nama**. This was completed in 1589 under the supervision of **master artist Daswant**. This manuscript was scribed in ornate calligraphy and contained 169 paintings.
 - **Artists like Govardhan and Miskin** were celebrated for their visuals of court scenes. Akbar Nama, an extraordinary manuscript, containing a detailed account of Akbar's political and personal life, was one of the most expensive projects undertaken by Akbar.
 - In most of the paintings, produced from the time that the Europeans were in contact with the court of Akbar, we can see an increasing preference for a category of naturalism adapted to complement the growing diversity in medieval India.
 - **Madonna and Child (1580)** done in opaque watercolor on paper is an important early work of the Mughal School of Painting in this context. Madonna, here, is an extraordinary theme, which brings Byzantine art, the European classical, and its Renaissance to the Mughal atelier, where it is translated and transformed into an entirely different visual experience. Virgin Mary is draped in a classical manner. The attachment displayed between the mother and the child was inspired by the humanist interpretation in European Renaissance art. The physiology of the child, and certain details like the fan, and jewelry completely integrated the work to an Indian milieu.

Q 39.B

- **Recent context:** The first two cases of the Marburg virus disease, a highly infectious Ebola-like disease, have been confirmed officially by Ghana after test results were verified by a Senegal laboratory.
- It was first detected in 1967 after simultaneous outbreaks in **Marburg and Frankfurt in Germany**; and in Belgrade, Serbia. Due to its detection in Marburg, it got its name.
- According to WHO, Marburg virus disease (MVD), is a severe, often fatal hemorrhagic fever. **It was earlier known as Marburg haemorrhagic fever.**
- Marburg, like Ebola, is a filovirus; and both diseases are clinically similar. Marburg virus disease (MVD) is a rare but severe hemorrhagic fever which affects both people and non-human primates. MVD is caused by the Marburg virus, a genetically unique zoonotic (or, animal-borne) RNA virus of the filovirus family. **Hence statement 1 is correct.**
- **Rousettus fruit bats are considered the natural hosts for the Marburg virus. Hence statement 2 is not correct.**
- According to WHO, African green monkeys imported from Uganda were the source of the first human infection.
- The disease has an average fatality rate of around 50%. However, it can be as low as 24% or as high as 88% depending on virus strain and case management, says the WHO.

- It is difficult to clinically distinguish MVD from diseases such as malaria, typhoid fever and other viral haemorrhagic fevers. However, it is confirmed by lab testing of samples, which like Coronavirus and Ebola are extreme biohazard risks.
- There is no approved antiviral treatment or vaccine for MVD as of now.

Q 40.C

- The decline of the Pratihara dynasty in the 10th century CE opened the way for their own feudatory chiefs to declare themselves as independent powers. The **Chalukyas, Chandellas, Chahamanas, Gahadawalas, Paramaras, Kalachuris, and Guhilas**, all of whom were the feudatory chiefs of the Gurjara-Pratiharas in different regions, thus, became independent as distinct Rajput clans in their own territories.
- The **Gahadawalas** occupied Kannauj in the 11th century. From Kannauj they ruled over the major portions of the Gangetic doab during 1090-1193. The Gahadawala king Jayachandra is usually styled as the king of Benares by Muslim historians owing to his intimate connection with the city of Banaras.
- **Chahamanas** came into prominence after the decline of the Gurjara-Pratiharas. There were numerous branches of the Chahamanas but some of them were unquestionably the feudatories of Pratiharas of Avanti and Kannauj. It also remains a fact that during 750-950 CE most of the regions ruled by the Chahamanas formed part of Pratihara dominion. In 973 CE they became practically independent. The city of Ajayameru (modern Ajmer) founded by king Ajayaraja was their political center and seat of power.
- **The Chandellas** ruled over central India between the 10th-13th centuries CE. Their territory was known as Jejakhukti (modern Bundelkhand).
- Another contemporary Rajput political power – **the Paramaras** – emerged in the region of Gujarat, Malwa, and Southern Rajputana out of the bitter struggle between the Gurjara-Pratiharas and the Rashtrakutas. The territory ruled over by the Paramaras of Malwa included Malwa proper and the adjoining districts.
- **The Maukhari dynasty was a post-Gupta royal Indian dynasty that controlled vast areas of Northern India for over six generations. They earlier served as vassals of the Guptas and later of Harsha's Vardhana dynasty. The Maukharis established their independence at Kannauj, during the mid 6th century.**
- **Hence option (c) is the correct answer.**

Q 41.D

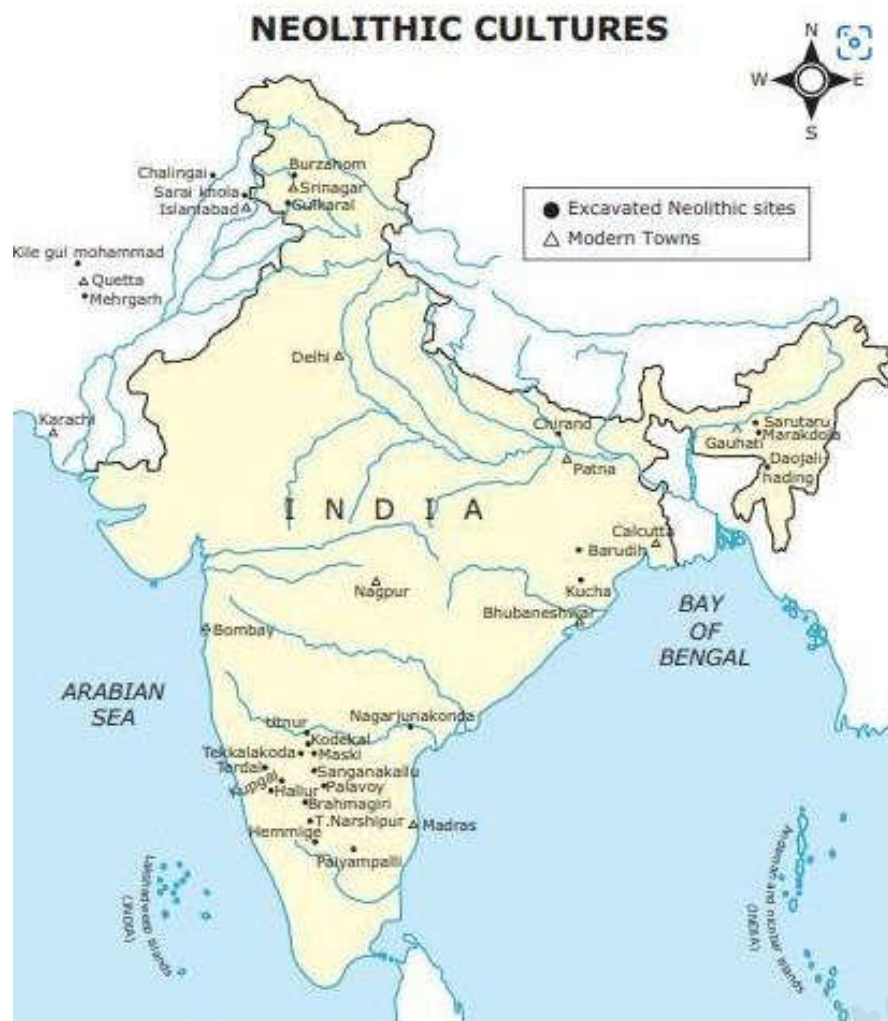
- Mughal Emperor Aurangzeb was the sixth emperor of the Mughal Empire and ruled from July 1658 until his death in 1707. In the Battle of Samugarh (1658), a war of succession to the Mughal throne, Aurangzeb defeated Dara and succeeded Shah Jahan as a Mughal ruler.
- Aurangzeb had the reputation of being an orthodox and god-fearing Muslim. Some of the features of his religious policies were:
- He re-imposed the payment of jizyah or poll tax for non-Muslims, which was abolished by Akbar.
- He discontinued the festival of Nauroz as it was considered a Zoroastrian practice favored by the Safavid rulers of Iran. **Hence statement 3 is not correct.**
- He banned the construction of new temples and the destruction of temples was not condoned but welcomed by him.
- **Muhtabis were appointed by Aurangzeb in all the provinces.** They were responsible for ensuring that things forbidden by the sharia and the zawabits (secular decree by Aurangzeb) were not openly flouted. Like their business was to see that wine and intoxicants were not consumed in public places. They were also responsible for regulating the houses of the ill repute, gambling dens, etc, and for checking weights and measures.
- He also forbade singing in the court and the official musicians were pensioned off. **Hence statement 1 is correct.**
- He discontinued the practice of jharokha darshan, started by Akbar, as he considered it to be a superstitious practice and against Islam. Similarly, he forbade the ceremony of weighing the emperor against gold and silver and other articles on his birthday.
- Jean-Baptiste Tavernier (1605 – 1689) was a 17th-century French gem merchant and traveler. Tavernier wrote a book "Travels In India" which contains detailed information about sea trade, sea routes, currency, weights and measures, and means of communication in India during the Mughal empire. He visited the Mughal court during Shahjahan and also stayed during the reign of Aurangzeb. **Hence statement 2 is correct.**

Q 42.D

- Recently Prime Minister Narendra Modi unveiled a 30-foot-tall bronze statue of Alluri Sitharama Raju at Bhimavaram in Andhra Pradesh as part of the year-long celebrations of the freedom fighter's 125th birth anniversary.
- Alluri Sitharama Raju was an Indian revolutionary involved in the Indian independence movement.
 - He was born in present-day Andhra Pradesh in 1897 or 1898.
 - He became a sanyasi at the age of 18, and gained a mystical aura among the hill and tribal peoples with his austerity, knowledge of astrology and medicine, and his ability to tame wild animals.
 - At a very young age, Raju channelled the discontent of the hill people in Ganjam, Visakhapatnam, and Godavari into a highly effective guerrilla resistance against the British.
 - Colonial rule threatened the tribals' traditional podu (shifting) cultivation, as the government sought to secure forest lands. The Forest Act of 1882 banned the collection of minor forest produce such as roots and leaves, and tribal people were forced into labour for the colonial government. While the tribals were subjected to exploitation by muttadars, village headmen commissioned by the colonial government to extract rent, the new laws and systems threatened their way of life itself.
 - Strong anti-government sentiment, shared by the muttadars who were aggrieved by the curtailment of their powers by the British, exploded into armed resistance in August 1922. Several hundred tribals led by Raju attacked the Chintapalle, Krishnadevipeta and Rajavommangi police stations in the Godavari agency.
 - Rising discontent towards the British led to the Rampa Rebellion/Manyam Rebellion of 1922, in which he played a major part as a leader.
 - ✓ The Rampa Rebellion coincided with Mahatma Gandhi's Non-Cooperation Movement. He persuaded people to wear khadi and give up drinking.
 - ✓ But at the same time, he asserted that India could be liberated only by the use of force, not non-violence.
 - He was nicknamed "**Manyam Veerudu**" (Hero of the Jungle) by local villagers for his heroic exploits.
 - In 1924, Raju was taken into police custody, tied to a tree, and shot by a public execution, effectively ending the armed rebellion.
 - **Hence option (d) is the correct answer.**

Q 43.A

- The Neolithic Age means the New Stone Age and the last part of the Stone Age. It spanned around 7000 BC. Neolithic was a very important stage of the history of human culture when humans were no longer dependent entirely on nature but had started to exploit nature to their own advantage.
- It indicates the transformation of society from a food gathering (hunting-gathering) economy to a food producing (agropastoral) economy.
- The Neolithic-agriculture-based regions (in Indian), can be categorized into four groups:
 - Indus system and its western borderland;
 - Ganga valley;
 - Western India and the northern Deccan;
 - and southern Deccan.
- **Important sites:**
 - Gufkral and **Burzahom** in Kashmir, Mahagara, Chopani Mando, and Koldihwa in Belan valley in Uttar Pradesh, and **Chirand** in Bihar.
- Following were the important sites in southern India:
 - **Kodekal**, **Utnur**, Nagatjunikonda, and Palavoy in Andhra Pradesh;
 - Tekkalkolta, Maski, Narsipur, Sangankallu, Hallur, and Brahmagiri in Karnataka;
 - **Paiyampalli** in Tamil Nadu. **Hence option (a) is the correct answer.**



Neolithic sites in India map

Q 44.C

- **Sri Vallabhacharya (c. 1479 – 1531 CE)** was a **devotional philosopher** who founded the **Pushtimarg sect in India**, following the **philosophy of Shuddha Advaita (Pure Non-dualism)**. Shuddhadvaita is the purely non-dual philosophy propounded by him. It was a Hindu Vaishnava tradition that focused on the worship of Krishna. **Hence statement 1 is not correct.**
- **Dadu Dayal (c. 1544-1603 CE)** was a significant representative of the **Nirguna Saint tradition in North India (Gujarat & Rajasthan)**. Nirgun is a form of aniconic worship of the formless as opposed to the saguna tradition, which is the worship of religious icons and deities with form. **Hence statement 2 is correct.**
 - **Dadu means brother**, and **Dayal means the compassionate one**; later, his followers came to be known as the **Dadupanthis**, who set up **ashrams known as Thambas**.
- **Surdas (c. 1483-1563 CE)** was a disciple of the Vallabhacharya. He was a blind poet whose songs were centered around Krishna. **His Sursagar recounts the exploits of Krishna during his childhood and youth with gentle affection and delightfulness. Hence statement 3 is correct.**

Q 45.C

- **Recent context: Prime Minister of India urged people to put the national flag as their display pictures on social media profiles as a part of Azadi ke Amrit Mahotsav.**
- The display of the national flag is governed by the provisions of the Emblems and Names (Prevention of Improper Use) Act, 1950 and the Prevention of Insults to National Honour Act, 1971. The Flag Code of India, 2002, is an attempt to bring together all such laws, conventions, practices and instructions for the guidance and benefit of all concerned.
 - The national flag shall be a tri-colour panel made up of three rectangular panels or sub-panels of equal widths. The colour of the top panel shall be India saffron (kesari) and that of the bottom panel shall be India green. **The middle panel shall be white, bearing at its centre the design of Ashoka Chakra in**

navy blue colour with 24 equally spaced spokes. The Ashoka Chakra shall preferably be screen printed or otherwise printed or stenciled or suitably embroidered and shall be completely visible on both sides of the flag in the centre of the white panel. **Hence statement 3 is not correct.**

- **The national flag of India shall be made of hand spun and hand woven wool/cotton/silk khadi bunting. Hence statement 2 is not correct.**
- The national flag shall be rectangular in shape. **The ratio of the length to the height (width) of the flag shall be 3:2. Hence statement 1 is correct.**

Q 46.D

- The Gupta period is remarkable for the production of secular literature, which consisted of a fair degree of ornate court poetry. **Bhasa** was an important poet in the early phase of the Gupta period and wrote thirteen plays. He wrote in Sanskrit, but his dramas also contain a substantial amount of Prakrit.
- Bhasa was the **author of a drama called Dradiracharudatta, which was later refashioned as Mrichchhakatika or the Little Clay Cart by Shudraka.** The play **deals with the love affair of a poor brahmana trader with a beautiful courtesan and is considered one of the best works of ancient drama. Hence option (d) is the correct answer.**
- In his plays Bhasa uses the term yavanika for the curtain, which suggests Greek contact. However, what has made the Gupta period particularly famous is the work of **Kalidasa** who lived in the second half of the fourth and the first half of the fifth century. He was the greatest poet of classical **Sanskrit literature and wrote Abhijnanashakuntalam** which is very highly regarded in world literature.
- It relates the love story of King Dushyanta and Shakuntala, whose son Bharata appears as a famous ruler. Shakuntalam was one of the earliest Indian works to be translated into European languages, the other work being the Bhagavadgita. The plays produced in India during the Gupta period have two common features.
- First, they are all comedies; no tragedies are found. Secondly, characters of the higher and lower classes do not speak the same language; women and Shudras featured in these plays use Prakrit whereas the higher classes use Sanskrit. We may recall that Ashoka and the Satavahanas used Prakrit as the state language.

Q 47.A

- **Recently, India contributed USD 2.5 million to the United Nations Relief and Works Agency for Palestine Refugees in the Near East (UNRWA).**
- **India is a dedicated donor to UNRWA,** having given USD 20 million in support of core UNRWA services to Palestine refugees across the Middle East since 2018. **Hence statement 3 is not correct.**
- Following the 1948 Arab-Israeli conflict, UNRWA was established by United Nations General Assembly Resolution 302 (IV) of 8 December 1949 to carry out direct relief and works programmes for Palestine refugees. The Agency began operations on 1 May 1950.
- **The agency was mandated to provide assistance and protection for about 5.6 million Palestinian refugees registered in the West Bank, Gaza Strip, and refugee camps in Lebanon, Syria, and Jordan. Hence statement 1 is correct.**
- The UNRWA services for Palestinian refugees involve education, health care, relief, infrastructure, camp improvement, protection and microfinance.
- **The United Nations Relief and Works Agency for Palestine Refugees (UNRWA) is funded almost entirely by voluntary contributions from UN Member States.** UNRWA also receives some funding from the Regular Budget of the United Nations, which is used mostly for international staffing costs. **Hence statement 2 is correct.**
- **The Agency's services encompass education, health care, relief and social services, camp infrastructure and improvement, microfinance and emergency assistance, including in times of armed conflict.**
- In the absence of a solution to the Palestine refugee problem, the General Assembly has repeatedly renewed UNRWA's mandate, most recently extending it until 30 June 2023.
- In January, the UNRWA announced that it needs 1.6 billion US dollars from the international community in 2022 to cover expenses and provide services and humanitarian development programs for Palestinian refugees.

Q 48.A

- The Vijayanagara Empire was based in the Deccan Plateau region of South India. It was established in 1336 by the brothers Harihara I and Bukka Raya I of the **Sangama dynasty.** The Sangama dynasty ruled between 1336 – 1485 CE. The Sangama dynasty was founded by Harihara I and Bukka. Their father had been taken prisoner in 1327 by Muhammad bin Tughluq. They founded Vijayanagara in 1336.

- **The Saluva dynasty ruled between 1485 – 1505 CE.** The Saluva dynasty was created by the Saluvas, who by historical tradition were natives of the Kalyani region of northern Karnataka in modern India.
- **The Tuluva dynasty ruled between 1491 – 1570 CE. Krishna Dev Raya** was an important ruler of this dynasty. Tuluva is the name of the third ruling dynasty of the Vijayanagara Empire. The dynasty traces its patrilineal ancestry to Tulu-speaking Nagavamshi Kshatriyas.
- **The Aravidu dynasty** ruled between 1542 – 1646 CE. Its founder was Tirumala Deva Raya, whose brother Rama Raya had been the masterful regent of the last ruler of the previous dynasty.
 - At Talikota in 1565, the ruler Rama Raya was killed. Though the Vijaynagar kingdom continued to exist for almost a hundred more years. Its size decreased and the Rayas no longer remained important in the politics of South India.
- **Hence option (a) is the correct answer.**

Q 49.C

- **After Skandgupta's reign, the Gupta power weakened.** Soon many subordinate rulers became independent. The later Gupta kings ruled between 475-530 CE. They ruled over small territories and functioned as petty rulers. **The Gupta rule finally came to end by the middle of the sixth century CE.** There were various reasons why the Gupta power weakened towards the end. A few of them can be summarised as mentioned below:
 - **Huna Invasion:** The successors of Chandragupta II had to face the Huna onslaught. Hunas were from Central Asia. They were a branch of the White Huns, the Hephthalites from Central Asia, known in the Indian sources as the Hunas. Before attacking the north-western Indian frontier, they had occupied Bactria and were threatening to cross the Hindu Kush mountains. Initially, the Gupta rulers were successful in fighting them back. However, the successors of Kumaragupta could not defend their kingdom. **Each wave of Huna invasions made the Gupta rule weaker. Hence statement 1 is correct.**
 - **Rise of Feudatories:** The Gupta Empire was further undermined by the rise of the feudatories. **The governors appointed by the Gupta kings in north Bengal and their feudatories in Samatata or south-east Bengal broke away from the Gupta control and started considering themselves as the central authority.** The later Gupthas of Magadha established their power in Bihar. Besides, the Maukharis rose to power in Bihar and Uttar Pradesh and had their capital at Kannauj. **Hence statement 2 is correct.**
 - **Land grants to Brahmins:** Historians have also mentioned that the **Guptas issued land grants to brahmana donees and in this process surrendered the revenue and administrative rights in favour of the donees.** This practice was bound to reduce revenues. **The migration of a guild of silk weavers from Gujarat to Malwa in 473 CE and their adoption of non-productive professions show that there was no great demand for the cloth produced by them. The advantages of the Gujarat trade gradually disappeared. Hence statement 3 is correct.**

Q 50.D

- **Recent Context: G7 leaders unveil \$600 billion plan to rival China's Belt and Road Initiative in developing nations. Hence statement 1 is not correct.**
- Partnership for Global Infrastructure and Investment (PGII) is a “values-driven, high-impact, and transparent infrastructure partnership to meet the enormous infrastructure needs of low and middle-income countries
 - And support the United States and its allies economic and national security interests.
- The US President, Joe Biden, and other G7 leaders have unveiled ambitious plans to **mobilise \$600 billion in funding by 2027 to deliver transparent and game-changing infrastructure projects in developing countries like India, in a move seen as a counter to China's Belt and Road Initiative. Hence statement 2 is not correct.**
- The U.S. President also announced the country's pledge to channel 200 billion USD in grants, public financing, and private capital over the next five years for the PGII.
- The European Commission President declared Europe's pledge of mobilizing 300 billion euros for the partnership over the same period.

Q 51.B

- **The most important literary source which mentions the Mauryan Empire is Megasthenes' Indica.** Megasthenes was a Seleukidian envoy who visited the Mauryan capital Pataliputra during the reign of Chandragupta Maurya. His account, Indica, encapsulates his impressions of India, particularly northern India, under Chandragupta Maurya. **Hence option 1 is correct.**

- **The other equally popular source is Kautilya's Arthashastra which explicitly mentions the Mauryan rule.** Traditionally Arthashastra is ascribed to Kautilya, also known as Vishnugupta or Chanakya, who is believed to have been Chandragupta's chief minister. He helped him overthrow the Nandas. **Hence option 2 is correct.**
- The texts like the **Divyavadana and the Ashokavadana as well as Sri Lankan Buddhist chronicles such as the Mahāvamsa and the Dipavamsa and the king lists in Puranas, again of a later period, refer to Mauryas.** Hence options 3 and 4 are correct.
- The **Divyāvadāna or Divine narratives is a Sanskrit anthology of Buddhist avadana tales.** This anthology explicitly mentions the rulers of the Mauryan Empire.
- **Dīpavaṃsa, the oldest extant historical record of Sri Lanka, was compiled in the 4th century.** It is considered to be one of the main sources drawn upon by the author of the later and more comprehensive historical chronicle of the Mahāvamsa. **In its emphasis on ecclesiastical (Buddhist) rather than political history and in the time span of its narrative,** the Dīpavaṃsa is similar to the Mahāvamsa. Unlike the Mahāvamsa, however, the Dīpavaṃsa is a crude, unpolished work—perhaps the first attempt of the Sinhalese people to write in Pāli, the sacred language of Buddhism. Because of its lack of organization and heterogeneity of style, it is generally considered the product of multiple authorship.

Q 52.A

- Rivalry with the Vijayanagar kingdom over the fertile Raichur doab, lying between the Krishna and Tungabhadra rivers, not only marked the early history of the Bahmani kingdom, but continued to be an enduring feature over two centuries. Bahman Shah had also to contend in the east with the rulers of Warangal and Orissa. In order to facilitate smooth administration, as followed in the Delhi Sultanate, he divided the kingdom into four territorial divisions called tarafs, each under a governor. Each governor commanded the army of his province (Gulbarga, Daulatabad, Bidar, and Berar) and was solely responsible for both its administration and the collection of revenue. The system worked well under a powerful king, but its dangers became apparent under a weak ruler. For the greater part of his reign of eleven years Bahman was engaged in subduing the unruly in his kingdom and in establishing order. His attempt to exact an annual tribute from the state of Warangal, the Reddi kingdoms of Rajahmundry and Kondavidu, led to frequent wars. Bahman Shah emerged victorious in all these expeditions and assumed the title Second Alexander on his coins.
- Most parts of the Deccan were conquered and annexed to the Delhi Sultanate during Muhammad Bin Tughluq's reign. He made elaborate administrative arrangements for the Deccan region. Ulugh Khan was appointed as the superior governor or "viceroy" of the region. The whole region was 'divided into **23 iqlims or provinces**. The most important of these were **Jajnagar (Orissa), Marhat (Maharashtra),** Telingana, Bidar, Kampili and Dwarsamudra.
- Subsequently, Malwa was also placed under the governor of the Deccan.
- Each iqlim was divided 'into a number of rural districts (shiq). Each shiq was divided into hazaris (one thousand) and sadis (one hundred) for collection of revenue. The main officers were shiqdars, wali, amiran-i hazarah and amiran-i sadah.
- **Hence option (a) is the correct answer.**

Q 53.D

- In parallel with the Bhakti movement in Hinduism, Sufism played a similar role in Islam. The terms Sufi, Wali, Darvesh and Fakir are used for Muslim saints who attempted to develop their intuitive faculties through ascetic exercises, contemplation, renunciation and self-denial. By the 12th century, Sufism had become an influential aspect of Islamic social life as it extended over almost the entire Muslim community.
- The Sufis were a class of philosophers remarkable for their religious catholicity and tolerance. Sufis regarded God as the supreme beauty and believed that one must admire it, take delight in His thought and concentrate his attention on Him only. They believed that God is 'Mashuq' (beloved) and Sufis are the 'Ashiqs' (lovers). Sufism crystallized into various 'Silsilahs' or orders. The most popular Sufi orders were Chistis, Suhrawardis, Qadiriya and Naqshbandis. Sufism took root in both rural and urban areas, and exercised a deep social, political and cultural influence on the masses. It rebelled against all forms of religious formalism, orthodoxy, falsehood and hypocrisy, and endeavoured to create a new world order in which spiritual bliss was the ultimate goal. At a time when struggle for political power was the prevailing trend, the Sufi saints reminded people of their moral obligations. In a world torn by strife and conflict they tried to bring peace and harmony.
- Sufism, also known as Tasawwuf, is a mystic body of religious practice, found mainly within Sunni Islam but also within Shia Islam which is characterized by a focus on Islamic spirituality, ritualism, asceticism and esotericism.

- **Tawakkul is a Sufism word that describes the concept of completely trusting in the plan of God. It means to have perfect faith in God. It symbolized the idea of observing containment. Hence pair 1 is correctly matched.**
- **Samāʿ, the sufi practice of listening to music and chanting to reinforce ecstasy and induce mystical trance. The Muslim orthodox regarded such practices as un-Islāmic, and the more puritanical among them associated the Sufi music, song, and dancing with drinking parties and immoral activities. Hence pair 2 is correctly matched.**
- **Shukr is an Arabic term which means thankfulness, gratitude, or acknowledgment by humans, and is a highly esteemed virtue in Islam. Hence pair 3 is correctly matched.**

Q 54.C

- **Alauddin Khalji was the first Sultan to have a large permanent standing army and paid them in cash from the royal treasury. An innovative Chehra and Daggh system was introduced by him wherein the Chehra (detailed description of each soldier) and Daggh (branding of horses) were maintained.**
- **Ala-ud-din's Internal Reforms: The vast annexation of territories was followed by extensive administrative reforms aimed at stabilising the government. Ala-uddin's first measure was to deprive the nobles of the wealth they had accumulated. It had provided them the leisure and means to hatch conspiracies against the Sultan. Marriage alliances between families of noble men were permitted only with the consent of the Sultan. The Sultan ordered that villages held by proprietary right, as free gift, or as a religious endowment be brought back under the royal authority and control. He curbed the powers of the traditional village officers by depriving them of their traditional privileges. Corrupt royal officials were dealt with sternly. The Sultan prohibited liquor and banned the use of intoxicating drugs. Gambling was forbidden and gamblers were driven out of the city. However, the widespread violations of prohibition rules eventually forced the Sultan to relax the restrictions.**
- **Ala-ud-din collected land taxes directly from the cultivators. The village headman who traditionally enjoyed the right to collect them was now deprived of it. The tax pressure of Alaud-din was on the rich and not on the poor. Ala-ud-din set up the postal system to keep in touch with all parts of his sprawling empire.**
- **To keep his soldiers satisfied with their salary the Khilji Sultan introduced strict price control measures based on production costs.**
 - **The growers were ordered to sell their grains in their fields at a fixed price and were not allowed to take any grain home for private sale. Every merchant was registered with the commerce ministry and had to sign a bond guaranteeing a regular supply of the goods they traded.**
 - **In order to restrict prices of essential commodities, Ala-ud-din set up an elaborate intelligence network to collect information on black-marketing and hoarding.**
- **Amīr Khusrau was a Sufi singer and often called the father of qawwali" (a devotional form of the singing of the Sufis in the Indian subcontinent). He was a disciple of Nizamuddin Auliya of Delhi. Alauddin Khalji patronized him along with Mir Hasan Delhvi as his court poet. Alauddin Khalji gave the title Tuti-i-Hind (Parrot of India) to Amir Khusrau.**
- **Hence option (c) is the correct answer.**

Q 55.B

- **During the second century BC to third century AD, many rulers from West Asia and Central Asia like the Indo-greeks, the Shakas, the Parthians, the Kushans assimilated into Indian culture and adopted Indian religions. Some adopted Vaishnavism, which means they worship Lord Vishnu, the God of protection and preservation. The Greek ambassador called Heliodorus set up a pillar in honour of Lord Vasudeva at Besnagar near Vidhisa in MP around the middle of the second century BC.**
- **A few other rulers adopted Buddhism. The famous Greek ruler Menander became a Buddhist. The questions and the answers that he exchanged with the Buddhist teacher Nagasena, also called Nagarjuna is a good source for the intellectual history of the post-Maurya period. The Kushan rulers worshipped both Shiva and Buddha and the images of these Gods appeared on the Kushan coins. Several Kushan rulers were worshippers of Vishnu, as was certainly the case with the Kushan ruler Vasudeva, whose very name is a synonym for Krishna, an incarnation of Vishnu.**
- **The Parthians were of Iranian origin and because of strong cultural connection with the Shakas, these groups were referred to in the Indian sources as Shaka-Pahlava. The important inscription indicating the Parthian rule in the northwestern area of Pakistan is the famous Takht-i-Bahi inscription recovered from Mardan near Peshawar. The inscription, dated in 45 AD, refers to Gondophernes or Gondophares as a Parthian ruler.**

- The Kushanas, originally belonged to western China. They are also called Yueh-chis. The Kushanas after defeating Shakas and Pahlavas created a big empire in Pakistan. The **first prominent ruler of the Kushana dynasty was Kujula Kadphises**. He was succeeded by his son Wema Kadphises. Next ruler was Kanishka. **Kanishka** is famous in history as a great patron of Buddhism. He convened the **fourth Buddhist Council** at Kundalavana (present-day Harwan near Srinagar in Jammu and Kashmir) in which a large number of Buddhist scholars took part. It was in this council that Buddhism got split into two schools – Hinayana and Mahayana.
- Hence, option (b) is the correct answer.

Q 56.B

- **Kinnal village in Karnataka is known for its unique variety of handcrafted toys and religious idols.**
 - **Kinnal (also spelled Kinhal) toys date back to the 15th and 16th centuries during which time it was patronized by the rich kings of the Vijayanagara empire and then later by the nawab of Koppal. Hence pair one is correctly matched.**
 - The exquisite carvings on the Hampi chariot as well as the famous mural paintings in the Pampapateshwara Temple are all believed to be the artwork of Kinhal craftsmen. Passed down from generations and having survived several centuries, **the wooden toys of Kinnal have been accorded a GI (Geographical Indication)** considering the fact that the craft is exclusive and native to this part of the state.
 - Kinnal idols known for their bright paint and vibrant colours are usually in the form of Gods, animals, wooden panels, murals and even masks.
 - The idols are handmade and the entire process is elaborate and quite a laborious task. The wood used in making the idols is of a tree locally known as “Ponki marra” that grows in and around Kinnal. The characteristic feature of this wood is that it is soft and lightweight and hence amenable to sculpting.
- **Etikoppaka is a small village on the banks of Varaha River at a distance of 64 kms away from the Visakhapatnam district of Andhra Pradesh. The name Etikoppaka is synonymous with the beautiful wooden artifacts and lacquer colours.**
 - **The toys are made with lacquer color and are traditionally known as Etikoppaka toys or Etikoppaka Bommalu. The village is very famous for its toys made of wood. The toys are also called lacquer toys because of the application of lacquer coating. Hence pair 2 is correctly matched.**
 - The toys are made out of wood and are coloured with natural dyes derived from seeds, lacquer, bark, roots and leaves. The wood used to make the toys is soft in nature and the art of toy making is also known as Turned wood Lacquer craft.
 - While making the Etikoppaka toys, lac, a colourless resinous secretion of numerous insects, is used. The already prepared vegetable dyes are further mixed to the lac, during the process of oxidation. After this process, the end product obtained is rich and colored lacquer. The lac dye is used for decorating the Etikoppaka toys, which are exported all over the world.
 - **Etikoppaka toys are granted GI tag status in India.**
- **Laiphadibi dolls are the traditional cloth dolls of the Meitei people of Manipur. Hence pair 3 is not correctly matched.**
 - Today, the tradition of making Laiphadibi at home has suffered a decline. The dolls made by professionals have striking features. The faces are mostly round, probably inspired by the notion that rounded faces are the epitome of beauty.
 - The eyes and brows are sewn with black thread, which is also used to simulate long luscious hair, and the nose and lips are stitched with red. The face is made of plain white cloth and the head stuffed with rags.
 - The outstretched hands and torso of the laidhibi are made of straw. These dolls have no legs and stand on their potloi (ornamental lower dress)



Q 57.C

- Maratha administration: Maratha kingdom consisted of territories known as the swarajya and the mughalai. In Swarajya his system of administration was implemented, while mughalai territory fell outside the swarajya and was subject to Maratha and other external raids.
- Central Government included:
 - Peshwa- Finance and general administration. Later he became the prime minister.
 - Sar-i-Naubat or Senapati – Military commander, an honorary post.
 - Amatya – Accountant General.
 - Waqenavis – Intelligence, posts and household affairs.
 - Sachiv – Correspondence.
 - Sumanta – Master of ceremonies.
- Local Government:
 - Most of the administrative reforms of Shivaji were based on the practices of the Deccan sultanates.
 - For the purpose of administrative convenience Shivaji divided his kingdom into four provinces and each was placed under a viceroy.
 - Shivaji abolished the system of granting jagirs and started the system of paying officers in cash.
 - Patel- village headman
 - Deshmukhs and deshpandes- Incharge of group of villages and supervised their functioning.
 - None of the officers enjoyed a hereditary status.
- The revenue system:
 - The revenue system of Shivaji was based on that of Malik Amber of Ahmadnagar.
 - He reduced the powers of the existing deshmukhs and kulkarnis. He appointed his own revenue officials called karkuns.
 - Lands were measured by using the measuring rod called kathi.
 - Lands were also classified into three categories: paddy fields, garden lands and hilly tracks.
 - The cultivators was at liberty to pay either in cash or kind, according to his own convenience and will.

- The amount of money to be paid to the state was fixed, which meant that there was not much scope for tax collectors to oppress the peasantry.
- Chauth and sardeshmukhi were the taxes collected not in the Maratha kingdom but in the neighbouring territories of the Mughal empire or Deccan sultanates.
- Chauth was one fourth of the land revenue paid to the Marathas in order to avoid the Maratha raids.
- Sardeshmukhi was an additional levy of ten percent on those lands which the Marathas claimed hereditary rights.
- To assist the king, there was a council of state ministers known as ashtapradhan:
 - Peshwa (Prime Minister): He was the head of both civil and military affairs.
 - **Mazumdar (auditor): He looked into the income and expenditure of the state.**
 - Wakis/Waqanavis: He was the incharge of king's private affairs.
 - Dabir: Foreign secretary
 - Surunavis (superintendent): He used to take care of all the official correspondences.
 - Pandit Rao: Ecclesiastical head
 - Senapati: Commander in chief
 - Nyayadhis: Chief Justice
- **Hence option (c) is the correct answer.**

Q 58.A

- **Humayun tomb:**
 - The grandness of Mughal architecture began with the construction of Humayun's tomb and **its design by Mirak Mirza Ghiyas from Persia**. He brought with him Persian craftsmen to work on the tomb. This tomb is the earliest specimen of a garden enclosure and is raised on an arcaded sandstone platform. It was **built in 1565 A.D.** nine years after the death of Humayun, **by his senior widow Bega Begam**. **Hence, statement 3 is correct.**
 - Inside the walled enclosure, the most notable features are the garden squares (Chahar bagh) with pathways and water channels, centrally located well proportional mausoleum topped by a double dome. There are several graves of Mughal rulers located inside the walled enclosure and from here in 1857 A.D; Lieutenant Hudson captured the last Mughal emperor Bahadur Shah II.
 - **It has an irregular octagon plan with four long sides and chamfered edges. Hence, statement 1 is correct.**
 - It is **surmounted by a 42.5 m high double dome** clad with marble flanked by pillared kiosks (chhatris) and the domes of the central chhatris are adorned with glazed ceramic tiles. **Hence, statement 2 is correct.**

Q 59.A

- Recently, NASA launched **Cislunar Autonomous Positioning System Technology Operations and Navigation Experiment (CAPSTONE Satellite)**. **Hence statement 2 is not correct.**
- It is a **microwave oven-sized CubeSat weighing just 55 pounds (25 kg)**.
 - It is designed to **test a unique, elliptical lunar orbit**.
 - It aims to help reduce risk for future spacecraft by validating innovative navigation technologies, and by verifying the dynamics of the halo-shaped orbit.
 - The orbit is known as a **near-rectilinear halo orbit (NRHO)**. **Hence statement 1 is correct.**
 - It is significantly elongated, and is located at a precise balance point in the gravities of Earth and the Moon.
 - It will gain experience with small dedicated launches of CubeSats beyond low-Earth orbit, to the Moon, and beyond, NASA said.
- **Mission objectives:**
 - Verify the characteristics of a cis-lunar near rectilinear halo orbit for future spacecraft.
 - Demonstrate entering and maintaining this unique orbit that provides a highly-efficient path to the Moon's surface and back.
 - Demonstrate spacecraft-to-spacecraft navigation services that allow future spacecraft to determine their location relative to the Moon without relying exclusively on tracking from Earth.
 - Lay a foundation for commercial support of future lunar operations.
 - Gain experience with small dedicated launches of CubeSats beyond low-Earth orbit, to the Moon, and beyond.

Q 60.D

- **Schools of Buddhism**
- **Mahayana:**
 - It is one of the two main schools of Buddhism.
 - The term Mahayana is a Sanskrit word that literally means "Great Vehicle".
 - It believes in the heavenliness of Buddha and Idol worship of Buddha and Bodhisattvas embodying Buddha Nature.
 - It originated in northern India and Kashmir and then spread east into Central Asia, East Asia, and some areas of Southeast Asia.
 - Buddhist schools embedded in China, Korea, Tibet, and Japan belong to the Mahayana tradition.
- **Hinayana**
 - Literally Lesser vehicle, also known as Abandoned Vehicle or Defective vehicle. It believes in the original teaching of Buddha or the Doctrine of elders.
 - It does not believe in Idol worship and tries to attain individual salvation through self-discipline and meditation.
 - Theravada is a Hinayana sect.
- **Theravada**
 - It is the most ancient branch of extant Buddhism today.
 - It remains closest to the original teachings of the Buddha.
 - Theravada Buddhism developed in Sri Lanka and subsequently spread to the rest of Southeast Asia. It is the dominant form of religion in Cambodia, Laos, Myanmar, Sri Lanka, and Thailand.
- **Vajrayana**
 - Vajrayana means **"The Vehicle of the Thunderbolt"**, also known as tantric Buddhism.
 - Established in Tibet in 11th CE
 - It believes that salvation can be attained by acquiring magical powers called **vajra**. **Hence, statement 1 is correct.**
 - It is grounded on esoteric elements and a very complex set of rituals compared with the rest of the Buddhist schools.
 - **Tara is the chief divinity** of this sect. **Hence, statement 2 is correct.**
 - It was popular **in eastern India**. **Hence, statement 3 is correct.**
 - Much importance is given to the role of the guru called Lama who has mastered the philosophical and ritual traditions. There is a long lineage of lamas. The Dalai Lama is a well-known Tibetan Lama.

Q 61.D

- **Sufi saint Khwaja Muin-ud-Din Chishti introduced the Chishti silsila in India.** He arrived in India in AD 1161 with Mahmud of Ghazni and found a base in Ajmer till AD 1236. **Other saints and followers of Khwaja Muin-ud-Din** were Bakhtiyar Kaki and his disciple Fariduddin Ganj-i-Shakar. **Hence statement 1 is not correct.**
- According to **Chisti Saint, selfless service is more critical than those obligatory prayers.** They advocate renouncing the word, **which means offering wealth and government service. This does not mean total withdrawal from society. Except for Nizamuddin Auliya, all the leading Chishti saints were married.** Thus, **married life for the saint was accepted as long as it did not come in the way of his ultimate spiritual life.** **Hence statement 3 is not correct.**
- **Khawja Mu'in al-Din Chishti settled in Ajmer because it was a small town away from the center of political activity, Delhi.** Khawja believed in the **spiritual life of seclusion; a small village was better than a large, bustling town.** Thus, his disciple, Hamiduddin, settled down at Nagpur. However, this does not mean that Chishti saints refrain from the state's service. **Hence statement 2 is not correct.**

Q 62.D

- The metaverse is an extension of our real world into the digital realm providing an immersive multi-user experience for anyone accessing it around the globe.
- CEOs like Mark Zuckerberg or Satya Nadella talk about it, the metaverse is the future of the internet. Or it's a video game. Or maybe it's a deeply uncomfortable, worse version of Zoom? It's hard to say.
- Facebook announced it was rebranding to Meta and would focus its future on the upcoming "metaverse." In the time since, what that term means hasn't gotten any clearer. Meta is building a VR social platform, Roblox is facilitating user-generated video games, and some companies are offering up little more than broken game worlds that happen to have NFTs attached.

- Accessing this virtual world requires the Internet and digital devices. **The technology behind this is called Augmented Reality (AR) and Virtual Reality (VR).**
- Applications of Metaverse:
- **E-Commerce: The metaverse can bridge the gap between the physical and virtual world, hence merging online and offline commerce into one.** This means that users will be able to experience the physical world via the metaverse, digitally, making online shopping more convenient. Companies would benefit hugely from the metaverse, as it not only widens their consumer base but also has the potential to get reviews on new products, thus pointing them in the right direction for the future. Brands can interact with the global audience through Metaverse in the ecommerce business framework despite the geological barriers.
- **Skill Enhancement: Metaverse can play a key role in skill enhancement in a remote manner** due to its scope of developing new experiential learning scenarios.
- **Virtual Tourism: With the 360° virtual tour,** the viewers can not only watch the location recorded but also be present digitally in the desired location with realistic effects. For instance, Virtual Reality Holiday “Try before you Fly” helps the potential tourists to visit their desired destinations virtually.
- **Education and Learning: VR combined with the effects of Metaverse has brought the learning experience to a qualitative new level.** Students can now watch live experiments with more intensive and high quality knowledge resources. Another Metaverse example is Mesh created by Microsoft which is a mixed reality platform where faculty, staff and students can interact using their 3D avatars.
- **Healthcare: Telemedicine and Telehealth is a concept fueled by the Metaverse post pandemic where patients and doctors can interact in virtual 3D clinics.**
- **Hence option (d) is the correct answer.**

Q 63.B

- Chavittu Natakam, a colorful and vigorous theatre form, is considered to be a folk art form noted for its attractive make-up of characters, elaborate costumes, detailed gestures, and well-defined body movements. It is believed that the art form flourished at Kodungalloor in Kerala with the spread of Christianity.
- **This 16th-century cosmopolitan art form derives its stories from the Holy Roman Empire, written in an early form of Malayalam.**
- **The drama was developed by the Portuguese missionaries to impart a cultural identity and teach the religious customs as approved by the Roman papacy to the newly converted Catholics.**
- **The influence of the western visual art opera can be discerned in Chavittu Natakam. Art forms like Kathakali and Kalaripayattu have also influenced Chavittu Natakam.**
- **The most attractive feature of Chavittu Natakam is the artists stamping the floor producing resonating sounds while dancing hence it is also called Stamping Drama.**
- This folk-drama dance takes place on a stage that is referred to as ‘thattu’. The ‘thattu’ is laid with planks of wood. The exquisite costumes of the artistes portray the characters on stage. **Generally, the costumes resemble ancient Greek-Roman soldiers and European kings.**
- Instruments like Chenda, Padathamber, Maddalam and Ilathalam, provides background music. These days Tabala, Fiddle, Flute and Bulbul are also played. The Chavittunatakam performance opens with an invocation and the opening sequence is in the form of a Virutham. It is a humming which is followed by a scene of a durbar and the play begins.
- **Hence option (b) is the correct answer.**
- **Naqal theatre art form is very popular in the villages of Punjab. It is enjoyable in as much as it presents subtly and sarcastically the seamy side of life.**
 - Those who specialized in this art were generally Mirasis, Naqaals and Bhands. A Naqaal troupe comprises, besides dancers and singers, clowns and musicians. The leader of the troupe is generally called Ustad. Legends and semi-historical tales like Dulla Bhatti, Sohni Mahiwal, Kima Malki, Hodi and Koklan form the popular repertoire of the Naqaal.
- **Koodiyaattam is one of the oldest traditional theatre forms of Kerala and is based on Sanskrit theatre traditions. The characters of this theatre form are Chakyaar or actor, Naambiyaar, the instrumentalists and Naangyaar, those taking on women's roles.**
 - The Sutradhar or narrator and the Vidushak or jesters are the protagonists. It is the Vidushak alone who delivers the dialogues. Emphasis on hand gestures and eye movements makes this dance and theatre form unique.
- **Padayani is a ritual theatre art form related to the Goddess temples in the Central Travancore regions of Kerala. Performed at night in the temple premises, the rituals of Padayani are a symbolic act for pleasing the Goddess Bhadrakali whose fury remains unabated even after her victory over the mythical**

demon, Darikan. The duration of the festivals varies from seven to 28 days according to the number of villages participating and organising it.

Q 64.C

- Adnyapatra, also pronounced as 'Ajnapatra', is a royal edict on the principles of Maratha policy written in Modi Marathi by Ramchandra Pant Amatya, who served on the Council of 8 (Ashta Pradhan) as the Finance Minister (Amatya) to Maratha King Shivaji, with the intention to guide Shivaji's grandson Sambhaji II. It is supposed to be the formal documentation of Shivaji's ideals, principles, and policies of state administration.
- Adnyapatra is a script of about 7000 words in Modi Marathi. It is divided into two sections. The first section comprises the first two chapters which give a brief narration of the achievements of Shivaji and his sons in building and preserving the Maratha Empire. The second section comprises seven chapters in which he discusses the principles of state policy and various aspects of administration he had learned while working with Shivaji.
- Modi was a script used to write Marathi for an estimated 700 years. The script was used for war correspondence and secret socio political conversations between Marathas.
- **Hence option (c) is the correct answer.**

Q 65.B

- The most important of the native successors of the Mauryas in the Deccan and Central India were the Satavahanas. The Satavahanas are considered to be the same as the Andhras mentioned in the Puranas.
- The Satavahanas **originally seems to have been a Deccan tribe. They however were so brahmanized that they claimed to be brahmanas.** Their most famous king, Gautamiputra Satakarni, described himself as a brahmana and claimed to have established the fourfold varna system which had fallen into disorder.
- The **Satavahanas were also the first rulers to make land grants to brahmanas**, although there are more instances of such grants being made to Buddhist monks.
- The most interesting social detail about the Satavahanas relates to their family structure. The **Satavahanas show traces of a matrilineal social structure. It was customary for their king to be named after his mother. Such names as Gautamiputra and Vashishtiputra** indicate that in their society the mother enjoyed a great deal of importance. Sometimes an inscription is issued both under the authority of the king and his mother. **Hence statement 2 is correct.**
- The Satavahana kingdom had three grades of feudatories. The highest grade was formed by the king who was called raja and had the right to strike coins. The second grade was formed by the mahabhoja and the third grade by the senapati. It stems that these feudatories and landed beneficiaries enjoyed some authority in their respective localities.
- **Satavahanas represented the march of triumphant Brahmanism. From the very outset, kings and queens performed such Vedic sacrifices as Ashvamedha, and vajapeya paying liberal sacrificial fees to the Brahmanas.** They also worshipped a large number of Vaishnava gods such as Krishna and Vasudeva. **Hence statement 1 is not correct.**
- **The official language of Satavahanas was Prakrit.** All inscriptions were composed in this language and were written in Brahmi script. **Hence statement 3 is correct.**

Q 66.B

- **Silambam is a popular variant of martial arts which is performed with the aid of some specific weapons, primarily bamboo sticks. It is widely performed in the regions of Tamil Nadu (India) which is also the birthplace of the game. The Tamil Sangam literature has the early records of Silambam. Hence pair 1 is correctly matched.**
 - The primary focus of the fighters is to use the bamboo stick as a weapon to defeat the opponents. The length of the weapon stick depends on the height of the fighters. In the battle, the fighter holds the weapon by stretching the arms around 3-quarters full.
 - In the era of 1760-1799, during Pulidevan's and Veera Pandiya Kattabomman's time, the art was reformed to fight against the Britishers. Post that, Silambam went popular throughout the country and its native state. But being a furious act, the Britishers banned the sport. The ban was lifted in India after independence.
- **Gatka is a traditional form of martial arts, which is historically associated with Sikh Gurus and is popular among the masses as a Sikh martial art.**
 - **It is a style of stick fighting between two or more practitioners, with wooden sticks intended to simulate swords. Hence pair 2 is correctly matched.**

- It was one of the basic techniques of self-defense for Sikh warriors during the martial period of the great Sikh Gurus.
- Creating an amalgamation between acrobatics and swords fight, Gatka was introduced as a means of warfare when the Sikhs were fighting the Mughal Empire during the late 17th century.
- **Thoda is a martial art form of Himachal Pradesh that relies on one's archery prowess. This martial art has its origin in Kullu. Hence pair 3 is not correctly matched.**
 - Thoda, the name is derived, from the round piece of wood fixed to the head of the arrow, which is used to blunt its wounding potential.
 - The equipment required for Thoda are bows and arrows. Wooden bows measuring 1.5m to 2m, to suit the height of the archer and wooden arrows in proportion to the length of the bow, are prepared by skilled and traditional artisans.

Q 67.D

- **Coins of the Maurya period are without legends. Known as karshapana coins, they do not specify the issuing authority; they do carry certain symbols that have been associated with Mauryan kings. Punch-marked coins, mostly of silver, were issued during the Mauryan period. Hence statement 1 is not correct.**
- **The punch-marked coins of the Mauryas contain uniform symbols.** Most probably, they were issued by the central authority. These symbols include crescent-on-arches, peacock-on-arches, etc.
- The Mauryan coinage **consisted almost exclusively of silver karshapanas** of roughly 3.4 gm, a series that continued the Magadha karshapana series. **The gold coins were not used during the Mauryan Empire, and it was the Kushana Rulers who issued coins of gold at a very great magnitude. Hence statement 2 is not correct.**

Q 68.B

- Akbar used different kinds of land revenue and land measurement systems to improvise the older systems and enhance the revenue without hurting the peasants.
- In 1580, Akbar instituted a new land revenue system called dahsala. Under this system, the average produce of different crops as well as average prices prevailing over the last ten (dah) years were calculated. One-third of the average share was state's share.
- **Akbar also followed the old batai or ghalla-bakshi system.** In this system, the produce was divided between peasants and the state in a fixed proportion. The crop was divided after it has been thrashed, or when it has been cut and tied in stacks, or while it was standing in the field.
- **A third system used under Akbar's reign was nasaq.** It meant a rough calculation of the amount payable by the peasant on the basis of what he has been paying in the past.
- **Hence option (b) is the correct answer.**

Q 69.D

- **Pongal**, is a major festival celebrated for continuous four days in **Andhra Pradesh and Tamil Nadu**. During this festival, every house of Andhra Pradesh is beautifully decorated with rangoli and flowers. Preparing the traditional dish Pongal on this festival is a must.
- **Ugadi** is the first day of Hindu Lunisolar calendar which is officially known as **New year day for Andhra Pradesh**. Ugadi pacchadi, a dish made of sweet, bitter and sour flavours is the main highlight of the festival. Colloquially known as Chandramana Ugadi, this auspicious day falls in March and April month.
- **Chettikulangara Bharani** is one of the spectacular festivals celebrated at Chettikulangara Temple near Mavelikara in **Alappuzha, Kerala**. Held during the Malayalam month of Kumbham (February-March), the festival is dedicated to Goddess Bhagavathy. The entire town comes to life and the merriment covers its landscape. The primary attractions during the festival are the Kettukazcha. These are giant structures that are decorated with cloth, flowers, and ornaments. It is an important moment for the 13 'karas' or regions that comprise the area.
- **Bonalu is an annual festival celebrated in twin cities Hyderabad and Secunderabad, and parts of Telangana.** The festival is dedicated to **Goddess Mahakali**, as per belief, a ferocious avatar of Goddess Durga.
 - The word "**Bonam**" signifies a meal or a feast in **Telugu**.
 - During this, the devotees offer rice cooked with milk and jaggery in an earthen or brass pot, which is decorated with neem leaves, turmeric and vermilion. Women carry these pots on their heads to temples.
 - Apart from Yellamma, during the Bonalu festival, the different forms of Mahakali are worshipped, including Mysamma, Dokkalamma, Pedamma, Pochamma, Yellamma, Poleramma and Ankamma. **Hence option (d) is the correct answer.**

Q 70.B

- In the south, **Dantidurga** was the founder of the dynasty called, the Rashtrakuta dynasty (8th AD).
- The capital of the Rastrakutas was **Manyakheta or Malkhed near Sholapur**.
- It was under the king Dhruva that the Rashtrakutas turned towards north India in a bid to control Kannauj, then the imperial city and it led to the beginning of the 'Tripartite struggle'.
- One of the important kings of the Rashtrakuta dynasty was **Krishna I**. He built the famous **Kailasha temple at Ellora** (near Aurangabad, Maharashtra). It is dedicated to **Lord Shiva and is monolithic** i.e. made of one single piece of rock. **Hence, statement 2 is correct.**
- The Arab accounts inform us that the Rashtrakutas were quite friendly with the Arab traders who visited their empire. These traders were allowed to **build mosques and follow their religion without any hindrance**. It testifies to the liberal attitude of the Rashtrakuta kings and also to their desire to draw economic benefit from the growing sea trade conducted by the Arabs at that time. **Hence, statement 1 is not correct.**
 - The Rashtrakuta kings presented an excellent example of religious tolerance and gave patronage to all the faiths such as Hinduism, Jainism, Buddhism, and Islam. They did not kill anyone for his or her religious beliefs. Toleration was also shown between the members of the different sects. The hope that religious charity will earn divine favour became a funding source to religious and socio-religious activities.

Q 71.A

- Though there was the rise of feudatories during the Gupta times but Gupta empire had an elaborate administrative system that was in operation in areas that were directly controlled by them. The King remained the central figure of administration and he was assisted by the Council of Ministers and other officials. The Guptas must have had a big army organization. At the time of war, the King led his army but ordinarily, there was a minister called 'Sandhi-Vigrahika' (Minister in charge of peace and war) who was helped by a group of high officials. The official title Mahabaladhikrita occurs in many inscriptions. **Officials like Pilupati (head of elephants), Asvapati (head of horses), and Narapati (head of footsoldiers) worked under Sandhi-Vigrahika.** The army was paid in cash and its needs were well looked after by an officer-in-charge of stores called Ranabhandagarika. Amongst other duties, this officer was to look after the supply of offensive and defensive weapons such as battle-axes, bows, and arrows, spear pikes, swords, lances, javelins, etc. **Hence, option (a) is the correct answer.**

Q 72.B

- **Philosophy in Medieval India**
 - The major religious movements were brought about by the mystics. They contributed to the religious ideas and beliefs. Bhakti saints like Vallabhacharya, Ramanuja, Nimbaraka brought about new philosophical thinking which had its origin in Shankaracharya's advaita (non-dualism) philosophy.
 - **Vishistadvaita of Ramanujacharya:** Vishistadvaita means modified monism. The ultimate reality according to this philosophy is Brahman (God) and matter and soul are his qualities.
 - **Sivadvaita of Srikanthacharya:** According to this philosophy, the ultimate Brahman is Shiva, endowed with Shakti. Shiva exists in this world as well as beyond it.
 - **Dvaita of Madhavacharya:** The literal meaning of dvaita is the dualism that stands in opposition to the non-dualism and monism of Shankaracharya. He believed that the world is not an illusion (maya) but a reality full of differences. **Hence pair 3 is correctly matched.**
 - **Dvaitadvaita of Nimbaraka:** Dvaitadvaita means dualistic monism. According to this philosophy, God transformed himself into the world and soul. This world and soul are different from God (Brahman). They could survive with the support of God only. They are separate but dependent. **Hence pair 1 is correctly matched**
 - **Suddhadvaita of Vallabhacharya:** Vallabhacharya wrote commentaries on Vedanta Sutra and Bhagavad Gita. For him, Brahman (God) was Sri Krishna who manifested himself as souls and matter. God and soul are not distinct, but one. The stress was on pure non-dualism. His philosophy came to be known as Pushtimarga (the path of grace) and the school was called Rudrasampradaya. **Hence pair 2 is not correctly matched**

Q 73.C

- During the 15th century, the Persian style of painting started influencing the Western Indian style of painting as is evident from the Persian facial types and hunting scenes appearing on the borders of some of the illustrated manuscripts of the Kalpasutra. The introduction of the use of ultramarine blue and gold colour in the Western Indian manuscripts is also believed to be due to the influence of the Persian painting.

- Some of the finest examples of painting belonging to the first half of the 16th century however are represented by a group of miniatures generally designated as the "Kulhadar Group".
- This group includes illustrations of the 'Chaurapanchasika' - "Fifty Verses of the Thief by Bilhan, the Gita Govinda, the Bhagavata Purana and Ragamala.
 - The style of these miniatures is marked by the use of brilliant contrasting colours, vigorous and angular drawing, transparent drapery and the appearance of conical caps 'Kulha' on which turbans are worn by the male figures.
 - Chaurapanchasika stands out to become the marker of age. It was a secular tale of Champavati and Bilhana a very popular tale of North India.
- Hence option (c) is the correct answer.

Q 74.B

- Remnants of rock paintings have been found on the walls of the caves situated in several districts of Madhya Pradesh, Uttar Pradesh, Andhra Pradesh, Karnataka and Bihar.
- Some paintings have been reported from the Kumaon hills in Uttarakhand also. The rock shelters on banks of the River Suyal at Lakhudiyar, about twenty kilometres on the Almora– Barechana road, bear these prehistoric paintings.
 - **Lakhudiyar literally means one lakh caves.**
 - The paintings here can be divided into three categories: man, animal and geometric patterns in **white, black and red ochre.**
 - **Humans are represented in stick-like forms.**
 - A long-snouted animal, a fox and a multiple legged lizard are the **main animal motifs.**
 - Wavy lines, rectangle-filled geometric designs, and groups of dots can also be seen here.
 - One of the interesting scenes depicted here is of **hand-linked dancing human figures.**
 - There is some superimposition of paintings. The earliest are in black; over these are **red ochre paintings** and the last group comprises white paintings.
- Hence option (b) is the correct answer.

Q 75.A

- India's bioeconomy is likely to touch USD 150 billion by 2025 and over USD 300 billion by 2030, a report said on Tuesday.
- **The India BioEconomy Report 2022, based on the data on biotech sector's contribution to the economy, has been brought out by the Biotechnology Industry Research Assistance Council (BIRAC).**
- The report said the country's bioeconomy has reached over USD 80 billion in 2021, registering a 14.1 per cent growth over USD 70.2 billion in 2020.
- **Biotechnology Industry Research Assistance Council (BIRAC)** is a not-for-profit Section 8, Schedule B, Public Sector Enterprise, set up by Department of Biotechnology (DBT), Government of India. BIRAC is a industry-academia interface and implements its mandate through a wide range of impact initiatives, be it providing access to risk capital through targeted funding, technology transfer, IP management and handholding schemes that help bring innovation excellence to the biotech firms and make them globally competitive.
- **Biotechnology Sector in India:** India is among the top 3 in South Asia and top 12 destinations for biotechnology in the world, with approximately 3% share in the global Biotechnology industry. Moreover, India has 2nd highest number of USFDA approved manufacturing plants outside the US. This sunrise sector enables technology led solutions for Healthcare, Industrial manufacturing, Agriculture, Environment and Clean Energy.

Q 76.B

- Though agriculture was the mainstay of the economy during the Gupta period, there were other occupations as well such as commerce and the production of crafts. Different social groups were engaged in these occupations.
- **Certain changes were witnessed in the agrarian society at this time. The land system became more complex. Grants of land were being made to religious and ritual specialists or to officers.** In those cases when the land granted to Brahmanas was a wetland or forest, the donees took on the role of a pioneer in introducing agriculture. The Brahmanas became proficient in supervising agrarian activities, helped by manuals on agriculture, such as Krishiparashara, which may be dated to this or the subsequent period. **In the Gupta period, land taxes increased in number, and those on trade and commerce decreased. Hence statement 1 is not correct.**

- Probably the king collected taxes varying from one-fourth to one-sixth of the produce. In addition, **whenever the royal army passed through the countryside, the local people had to feed it.** The peasants had also to supply animals, food grains, furniture, etc., for the maintenance of royal officers on duty in rural areas. In central and western India, **the villagers were also subjected to forced labour called *vishti* by the royal army and officials.** Hence statement 2 is correct.
- For the smooth function of transactions in the economy, various types of coins were issued by different rulers. In ancient India, **the Guptas were known for issuing a good number of gold coins, which were called dinars in their inscriptions.** However, in gold content, the Gupta coins are not as pure as the Kushana ones and most probably not used by ordinary peoples in their daily transactions. Daily transactions mostly were done in copper coins or other modes of currency. But in contrast to those of the Kushanas, the Gupta copper coins are very few. Thus, it seems that commercial activities in this period were not as flourishing as they were in the Kushana period. **Hence statement 3 is not correct.**

Q 77.A

- **Ashoka was converted to Buddhism as a result of the Kalinga war.** According to tradition, he became a monk, made huge gifts to the Buddhists, and undertook pilgrimages to the Buddhist shrines. **According to tradition, Ashoka held the third Buddhist council (sangiti)** and missionaries were sent not only to south India but also to Sri Lanka, Myanmar (Burma), and other countries to convert the people there.
- **Ashoka appointed dhammamahamatras to propagate dharma among various social groups, including women, and appointed rajukas for the administration of justice in his empire.**
- Ashoka disapproved of rituals, especially those observed by women. **He forbade killing certain birds and animals, prohibited the slaughter of animals in the capital, and forbade the slaughter of animals in sacrifices.** He banned gay social functions in which people indulged in an excess of revelry. **Hence statement 1 is correct.**
- His teachings were meant to strengthen the institution of family and the existing social classes. He held that if the people behaved well they would go to heaven, **but never said that they would attain nirvana, which was the goal of Buddhist teachings.**
- He ordained that people should obey their parents, pay respect to the Brahmanas and Buddhist monks, and show mercy to slaves and servants. Above all, the dhammalipi asks the people to show firm devotion (dridha bhakti) or loyalty to king. **These instructions are found in both the Buddhist and brahmanical faiths. He did not try to foist his Buddhist faith on his subjects, rather he made gifts to Non -Buddhist and even anti-Buddhist sects. Hence statement 2 is not correct.**

Q 78.D

- **Mahayana Buddhism:**
 - It means the greater vehicle.
 - **The school is more liberal and believes in the heavenliness of Buddha and Bodhisattvas embodying Buddha Nature. Hence statement 2 is correct.**
 - **The ultimate goal under Mahayana is “spiritual upliftment”. The Mahayana followers believe in idol or image worship of Buddha. Hence statement 1 is correct.**
 - The concept of Bodhisattva is the result of Mahayana Buddhism. Mahayana is also called “Bodhisattvayana”, or the “Bodhisattva Vehicle. That is to say, the followers believe in Bodhisattva concept of the salvation of all conscious individual. In other words, they believe in universal liberation from the suffering of all beings.
 - A bodhisattva seeks complete enlightenment for the benefit of all beings. A bodhisattva who has accomplished this goal is called a Samyaksambuddha.
 - Prominent Mahayana texts include Lotus Sutra, Mahavamsa, etc.
 - As per Lotus Sutra, the Mahayana school believes in **six perfections (or paramitas)** to be followed by an individual:
 - ✓ **Dana** (generosity)
 - ✓ **Sila** (virtue, morality, discipline and proper conduct)
 - ✓ **Kṣanti** (patience, tolerance, acceptance)
 - ✓ **Virya** (energy, diligence, vigor, effort)
 - ✓ **Dhyana** (one-pointed concentration)
 - ✓ **Prajna** (wisdom and insight)
 - As per scholars, one of the sub-sects of Mahayana being developed in the later periods was Vajrayana.
 - **The Mahayana scholars predominantly used Sanskrit as a language. Hence statement 3 is not correct.**

- **Emperor Kaniska** convened the **fourth great Buddhist council** in Kashmir, which marked the beginnings of Mahayana Buddhism (as Hinayana and Mahayana sects of Buddhism diverged after this council). He patronized the new Mahayana sect of Buddhism which led to its popularity. **Hence statement 4 is correct.**
- Presently, the majority of the Buddhist followers in the world belong to Mahayana sect. (around 53.2% as per 2010 report)
- Other countries following it includes Nepal, Bangladesh, Japan, Vietnam, Indonesia, Malaysia, Singapore, Mongolia, China, Bhutan, Tibet, etc.

Q 79.C

- **Department of Pharmaceuticals, Ministry of Chemicals and Fertilizers has released the guidelines for the scheme “Strengthening of Pharmaceutical Industry (SPI)”**, with a total financial outlay of Rs.500 Cr for the period from FY 21-22 to FY 25-26.
- **The scheme will address the rising demand in terms of support required to existing Pharma clusters and MSMEs across the country to improve their productivity, quality and sustainability.**
- The objectives of the scheme “Strengthening of Pharmaceutical Industry (SPI) are to strengthen the existing infrastructure facilities in order to make India a global leader in the Pharma Sector.
- Under the Scheme, financial assistance to pharma clusters will be provided for creation of Common Facilities. This will not only improve the quality but also ensure the sustainable growth of clusters.
- Further, in order to upgrade the production facilities of SMEs and MSMEs so as to meet national and international regulatory standards (WHO-GMP or Schedule-M), interest subvention or capital subsidy on their capital loans will be provided, which will further facilitate the growth in volumes as well as in quality.
- **The Scheme has 3 components / sub-schemes:**
 - **Assistance to Pharmaceutical Industry for Common Facilities (APICF)**, to strengthen the existing pharmaceutical clusters’ capacity for their sustained growth by creating common facilities;
 - **Pharmaceutical Technology Upgradation Assistance Scheme (PTUAS)** to facilitate Micro, Small and Medium Pharma Enterprises (MSMEs) of proven track record to meet national and international regulatory standards and
 - **Pharmaceutical & Medical Devices Promotion and Development Scheme (PMPDS)** to facilitate growth and development of Pharmaceutical and Medical Devices Sectors through study/survey reports, awareness programs, creation of database, and promotion of industry.
- **Pradhan Mantri Bhartiya Janaushadhi Pariyojana (PMBJP):** Pradhan Mantri Bhartiya Janaushadhi Pariyojana (PMBJP) is a campaign launched by the Department of Pharmaceuticals to provide quality medicines at affordable prices to the masses. PMBJP stores have been set up to provide generic drugs, which are available at lesser prices but are equivalent in quality and efficacy as expensive branded drugs. It was launched by the Department of Pharmaceuticals in November 2008 under the name Jan Aushadi Campaign. Pharmaceutical & Medical Devices Bureau of India (PMBI) is the implementation agency for PMBJP. **It is not a sub-scheme under SPI.**
- **The National Pharmaceutical Pricing Authority (NPPA)**, Department of Pharmaceuticals implements the Scheme named **“Consumer Awareness, Publicity and Price Monitoring (CAPPMP)” Scheme**. The Scheme has two components viz., (a) National component and (b) State component.
 - The national component covers the expenditure for publicity through print and electronic media, organizing seminars for consumer awareness, purchase of samples etc.
 - Under the State Component of the Scheme, it is planned to set up Price Monitoring and Resource Units (PMRUs) in the States. PMRU is a registered society under the chairmanship of the State Drug Controller. The representatives of NPPA/State Health Department, civil societies and other stakeholders are members of the PMRU.
- **Hence option (c) is the correct answer.**

Q 80.D

- Babur introduced a **new warfare method known as the Tughlaq system**, in the first battle of Panipat. Tughlaq meant dividing the whole army into various units, viz. the Left, the Right, and the Centre.
 - The Left and Right divisions were further subdivided into Forward and Rear divisions. The Centre Forward division was then provided with carts (araba) which were placed in rows facing the enemy and tied to each other with animal hide ropes.
 - Behind them were placed cannons protected and supported by mantelets which could be used to easily maneuver the cannons.

- **This introduction of a new mode of warfare** led to the **popularization of gunpowder, cavalry, and artillery in India**. However, **gunpowder was known earlier in India**, but its **use for artillery became common in north India with the advent of Babur**. Hence option (d) is the correct answer.
- **Facts about the other three battles:**
 - **Battle of Samugarh (1658):** The battle of Samugarh was fought between Shahjahan's sons Dara Shikoh (the eldest son and heir apparent) and his two younger brothers Aurangzeb and Murad Baksh (third and fourth sons of Shah Jahan) to decide who will be the heir of the throne after their father.
 - **Battle of Haldighati (1576):** The Battle of Haldighati was a battle fought on 18 June 1576 between the Mewar forces led by Maharana Pratap and the Mughal forces led by Man Singh I of Amber. Akbar was the Mughal Emperor at the time.
 - **Sultana Chand Bibi:** Sultana Chand Bibi (1550–1599 CE) was an Indian ruler and warrior. She acted as the Regent of Bijapur Sultanate during the minority of Ibrahim Adil Shah II in 1580-1590, and regent of Ahmednagar Sultanate during the minority of her great nephew Bahadur Shah in 1595-1600. Chand Bibi is best known for defending **Ahmednagar against the Mughal forces of Emperor Akbar in 1595**.

Q 81.C

- **System of Administration under Gupta Empire:**
 - In contrast to the Maurya rulers, the Gupta kings **adopted pompous titles such as parameshvara, maharajadhiraja, and paramabhattacharaka** which signify that they ruled over many lesser kings in their empire. Kingship was hereditary, but royal power was limited by the want of firm adherence to primogeniture.
 - Evidently, the king maintained a standing army, which was supplemented by the forces occasionally supplied by his feudatories. Horse chariots receded into the background, and cavalry came to the fore. Horse archery became an important element in military tactics. **During the Gupta period, land taxes increased in number, and those on trade and commerce decreased.**
 - Probably the king collected **taxes varying from one-fourth to one-sixth of the produce**. In addition, whenever the royal army passed through the countryside, the local people had to feed it. The peasants had also to supply animals, food grains, furniture, etc., for the maintenance of royal officers on duty in rural areas.
 - The guilds of artisans, merchants, and others were governed by their own laws. Seals from Vaishali and from Bhita near Allahabad indicate that these guilds flourished during Gupta times.
 - The Gupta bureaucracy was not as elaborate as that of the Mauryas. The **most important officers in the Gupta empire were the Kumaramatyas**. They were appointed by the king in the home provinces and possibly paid in cash. **Hence pair 2 is not correctly matched.**
 - As the Guptas were possibly vaishyas, recruitment was not confined to the upper varnas only, but several offices were combined in the hands of the same person, and posts became hereditary. This naturally weakened royal control. The Guptas organized a system of provincial and local administration.
 - The **empire was divided into divisions called bhuktis, and each bhukti was placed under the charge of an Uparika**. The **bhuktis were divided into districts (vishayas), which were placed under the charge of a Vishayapati**. In eastern India, the **vishayas were divided into vithis**, which again were subdivided into villages. **Hence both pairs 1 and 3 are correctly matched.**
 - The **village headman gained importance in Gupta times**, managing village affairs with the assistance of elders. With the administration of a village or a small town, leading local elements were associated. No land transactions could be effected without their consent.

Q 82.D

- **Dhokra casting**
 - **Dhokra is an alloy of brass, nickel and zinc** which gives an antique look. The oldest specimen using such a technique was during the Harappan Period (famous **Dancing Girl of Mohenjo-Daro**).
 - Among the popular sculptural traditions, Dhokra or metal sculptures are **made from lost wax or cire perdue technique**. Hence, statement 1 is correct.
 - It is one of the most prominent **metal crafts of Bastar, Chhattisgarh, parts of Madhya Pradesh, Odisha and Midnapore in West Bengal**. It involves casting bronze through the lost wax method. Hence, statement 2 is correct.
 - The **metal craftsmen of Bastar are called ghadwa**. In popular etymology, the term 'ghadwa' means the act of shaping and creating. It is probably this that gives the casters their name. Hence, statement 3 is correct.

- Its motifs are mostly drawn from a folk culture which includes animal figures like elephants, horses, gods and goddesses, containers with lids, lamps and lamp stands, and intricate designs in the shape of trees and branches.
- **NOTE: Lost-wax process or the cire-perdue** is a method of metal casting in which molten metal is poured into a mould that has been created by means of a wax model. Once the mould is made, the wax model is melted and drained away. A hollow core is then filled with molten metal which takes the shape of the mould.

Q 83.C

- Chemistry in medieval India was closely associated with alchemy which was an integral part of the Tantric cult. Although the origin of alchemy in India may be traced to date as far back as that of the Atharva Veda, or even that of the Rig Veda, practical alchemy reached its peak only during the Tantric period.
- Tantric treatises, both Brahmanical and Buddhist, abound in recipes for such transmutation of base metals, particularly mercury into gold.
- **The book Rasaratnakara, written by the famous Buddhist alchemist Nagarjuna, contains descriptions of alchemical processes and preparations of many mercurial compounds. Hence statement 1 is correct.**
 - **It gives an account of many chemical processes like the extraction of zinc, mercury, and copper, and the preparation of crystalline red sulfide of mercury (svarnasindura or makaradhvaja). Hence statement 2 is correct.**
 - This medicament is still used as a panacea for many ailments by physicians in India following the indigenous system of medicine.
 - The treatise also describes more than two dozen varieties of apparatuses (yaniras) for carrying out various physico-chemical processes like distillation, sublimation, extraction, calcinations, digestion, evaporation, filtration, fumigation, fusion, pulverization, heating by steam and by sand, and the preparation of many metallic compounds.

Q 84.A

- Among the travellers whose writings on Vijayanagar have survived are **Niccolo de Conti**, a Venetian merchant (1420); **Abdur Razzaq Samarqandi**, a Persian scholar-envoy (1443); **Duarte Barbosa**, a Portuguese officer (c. 1515); and **Domingo Paes** (c. 1520) and **Fernao Nuniz** (1535–37), Portuguese merchant-adventurers.
 - The institution of nayaka was studied in detail by two Portuguese-**Fernao Nuniz and Domingo Paes**, who visited India during the reigns of **Krishnadeva Raya and Achyut Raya of the Tuluva dynasty** during the sixteenth century.
 - We get information about foreign trade from the Amuktamalyada of Krishnadeva Raya, Domingo Paes and Nuniz. They have given vivid descriptions of the horse trade.
 - Nicolò de' Conti did not write about his extensive travels. Our knowledge of him has been filtered through the works of two men to whom he recounted his adventures.
- **Niccolo Manucci** (1639-1717) was a Venetian traveller who toured different parts of India including Bengal and left behind a reliable description of the land and people. He wrote accounts of the **Mughal Empire**.
- **Other travellers who visited the Vijaynagar empire are:**
 - **Abdur Razzaq:** He was a Persian, Timurid chronicler and scholar who visited the Vijaynagar Kingdom at the time of Dev Raya II as an ambassador of Shah Rukh, the Timurid dynasty ruler of Persia.
 - **Athanasius Nikitin:** He was the first noteworthy traveller from Russia.
 - **Duarte Barbosa:** He was a Portuguese Writer, scrivener and explorer who had given a vivid account of Vijaynagar governance under Krishna Deva Raya.
- **Hence option (a) is the correct answer.**

Q 85.A

- **Tholu Bommalata, a shadow theatre form of Andhra Pradesh has a rich tradition. In Tholu Bommalata the puppets are large in size and have jointed waist, shoulders, elbows and knees.**
 - They are coloured on both sides. Hence, these puppets throw coloured shadows on the screen. The music is dominantly influenced by the classical music of the region and the theme of the puppet plays are drawn from the Ramayana, Mahabharata and Puranas.
- **Yampuri is a traditional Rod puppet of Bihar.** These puppets are made of wood. Unlike the traditional Rod puppets of West Bengal and Orissa, these puppets are in one piece and have no joints. As these puppets have no joints, the manipulation is different from other Rod puppets and requires greater dexterity.

- **Pavakoothu is a traditional glove puppet play of Kerala. It came into existence during the 18th century due to the influence of Kathakali, the famous classical dance-drama of Kerala, on puppet performances.**
 - In Pavakoothu, the height of a puppet varies from one foot to two feet. The head and the arms are carved of wood and joined together with thick cloth, cut and stitched into a small bag.
 - The face of the puppets is decorated with paints, small and thin pieces of gilded tin, the feathers of the peacock, etc.
 - The manipulator puts his hand into the bag and moves the hands and head of the puppet. The musical instruments used during the performance are Chenda, Chengiloa, Ilathalam and Shankhathe conch.
 - The theme for Glove puppet plays in Kerala is based on the episodes from either the Ramayana or the Mahabharata.
- **Hence option (a) is the correct answer.**

Q 86.D

- **Delhi first become the capital of a kingdom under Tomar Rajput, who was defeated in the middle of the 12th Century by the Chauhan of Ajmer.** It was found under Tomaras and Chauhans that Delhi become an important commercial center. **Hence statement 1 is not correct.**
- **The Mongols under Genghis Khan invaded Transoxiana and North-East Iran in 1219 and the Delhi Sultanate faced their onslaught again soon after the reign of Alauddin Khaliji (reign: 1296-1316).** Their onslaught became more intense and ferocious during the Khilji rule. But Alauddin succeeded in crushing the Mongols. Shams ud-Din Iltutmish or Iltutmish was the third of the Mamluk kings who ruled the former Ghurid territories in northern India. His rule (1211-1236) preceded that of the Khiljis. **Hence statement 2 is not correct.**
- **Tughluq Dynasty (1320-1414) was also known as the Quaraunah Turks.** Ghiyassuddin Tughlaq was the founder/first ruler of the Tughlaq dynasty. **Hence statement 3 is not correct.**

Q 87.C

- **Recent context:** James Lovelock, environmental scientist who created Gaia ecology theory, passed away.
- About Gaia hypothesis (also known as Gaia theory or Gaia principle)
 - First proposed in the 1970s.
 - It **proposes that all organisms and their inorganic surroundings on Earth** are closely integrated to form a single and self-regulating complex system.
 - It postulates that this single system as a whole maintains Earth's surface in a habitable state by self-regulating feedback mechanisms.
- **Hence option (c) is the correct answer.**

Q 88.C

- The Akbar Nama was written to provide a detailed description of Akbar's reign in the traditional diachronic sense of recording politically significant events across time, as well as in the more novel sense of giving a synchronic picture of all aspects of Akbar's empire – geographic, social, administrative and cultural – without reference to chronology.
- Abu'l Fazl wrote in a language that was ornate and which attached importance to diction and rhythm, as texts were often read aloud. This Indo- Persian style was patronized at court, and there were a large number of writers who wanted to write like Abu'l Fazl. **Hence statement 1 is correct.**
- The Akbar Nama is divided into three books of which the first two are chronicles. The third book is Ain-i Akbari. The first volume contains the history of mankind from Adam to one celestial cycle of Akbar's life (30 years). The second volume closes in the forty-sixth regnal year (1601) of Akbar. The very next year Abu'l Fazl fell victim to a conspiracy hatched by Prince Salim and was murdered by his accomplice, Bir Singh Bundela.
- In the Ain-i Akbari, the Mughal Empire is presented as having a diverse population consisting of Hindus, Jains, Buddhists, and Muslims and a composite culture. Ain-i-Akbari embodies Ains or rules and regulations framed and put into effect for proper administration by Akbar. It is regarded as an administrative manual and is like a modern gazetteer. The regulations embodied in the Ain-i-Akbari provide information about Akbar's government, several departments, its different ranks, etc.
- Abul Fazl also discusses in this book the social condition, literary activities, and study of law and philosophy not only of Muslims but also of local Hindus, Jains, and other communities. There are, in addition, chapters on distinguished travelers, Muslim saints, and sufis. **Hence statement 2 is correct.**

Q 89.D

- After Ashoka's tryst with Dhamma and Buddhism, **the Sungas are known for having reverted to Brahmanical orthodoxy. In the Ayodhya inscription of Dhanadeva, Pushyamitra Sunga is credited with the performance of two Ashvamedha sacrifices. Buddhist sources claim that he persecuted the Buddhists.** Divyavadana depicts **Pushyamitra as a destroyer of Buddhist monasteries** and places of worship, particularly those constructed by Ashoka. For instance, it is said that he attempted to destroy the Kukuta Arama monastery at Pataliputra. **Hence statement 1 is not correct.**
- The **Ayodhya inscription of Dhanadeva praises Pushyamitra Shunga** having performed two horse sacrifices (dvirashvamedhayajin), which alludes to his military success. **It is the first inscription on stone or metal that mentions the name of Pushyamitra. Hence statement 3 is correct.**
- **Maharishi Patanjali, credited with the writing of the text called Mahabhasya** was a saint who is believed to have lived sometime during the 2nd century BCE. **He was a contemporary of Pushyamitra Shunga (185 – 149 BCE: the founder and first ruler of the Shunga dynasty). He was the priest in the Ashvamedha yajna performed by Pushyamitra Shunga. Hence statement 2 is correct.**

Q 90.C

- Recently, the Ministry of Road Transport and Highways approved the **draft GSR** notification to introduce **Bharat NCAP (New Car Assessment Program)**.
 - **Bharat NCAP (New Car Assessment Program)** will be rolled out from April 1, 2023.
 - It is a **new car safety assessment programme** which proposes a mechanism of awarding 'Star Ratings' to automobiles based upon their **performance in crash tests. Hence option (c) is the correct answer.**
 - Bharat NCAP standard is aligned with global benchmarks and it is beyond minimum regulatory requirement.
 - The proposed **Bharat NCAP** assessment will allocate Star Ratings from 1 to 5 stars.
 - The testing of vehicles for this programme will be carried out at testing agencies, with the necessary infrastructure.
 - It will be applicable on type approved motor vehicles of category M1 with gross vehicle weight less than 3.5 tonnes, **manufactured or imported** in the country.
 - ✓ M1 category motor vehicles are used for the carriage of passengers, comprising eight seats, in addition to the driver's seat.
 - Auto firms in India follow **AIS-145 (automotive Indian standard-145)**, which enforces safety features for vehicles such as seatbelts tell-tale, passenger airbags, and the speed limit alarm.
 - It will encourage manufacturers to **participate voluntarily** in the safety testing assessment programme and incorporate higher safety levels in new car models. It aims to **reduce 50 per cent road accident deaths by 2024.**
 - It will also promote a healthy competition among **original equipment manufacturers (OEMs)** in India to manufacture safer vehicles.
 - It will ensure **structural and passenger safety in cars**, along with increasing the export-worthiness of Indian automobiles.
 - It will prove to be a **critical instrument in making our automobile industry Aatmanirbhar** with the mission of making India the top automobile hub in the world.
- **Global NCAP:** The **US was the first country to start a programme** that provided information on car safety with regard to crashes to customers in 1978. Later, a number of similar programmes were started across regions.

Q 91.D

- Bharatnatyam Dance is considered to be over 2000 years old. Several texts beginning with Bharata Muni's Natya Shastra (200 B.C.E. to 200 C.E.) provide information on this dance form.
- There is also a great deal of visual evidence of this dance form in paintings and stone and metal sculptures of ancient times. On the gopurams of the Chidambaram temple, one can see a series of Bharatnatyam poses, frozen in stone as it were, by the sculptor.
- **Bharatnatyam dance is known to be ekaharya, where one dancer takes on many roles in a single performance. Hence statement 1 is correct.**
- In the early 19th century, the famous Tanjore Quartette, under the patronage of Raja Serfoji are said to have been responsible for the repertoire of Bharatnatyam dance as we see it today.
- The style was kept alive by the devadasis, who were young girls 'gifted' by their parents to the temples and who were married to the gods. The devadasis performed music and dance as offerings to the deities, in the temple courtyards.

- The repertoire of Bharatnatyam is extensive, however, a performance follows a regular pattern. **At first, there is an invocation song. The first dance item is the alarippu, literally meaning - to adorn with flowers.** It is an abstract piece combining pure dance with the recitation of sound syllables.
- **The next item, the jatiswaram is a short pure dance piece performed to the accompaniment of musical notes of any raga of Carnatic music.** Jatiswaram has no sahitya or words, but is composed of adavus which are pure dance sequences - nritta. They form the basis of training in Bharatnatyam dance.
- **As a solo dance, Bharatnatyam leans heavily on the abhinaya or mime aspect of dance - the nritya, where the dancer expresses the Sahitya through movement and mime. Shabdham follows the jatiswaram in a Bharatnatyam dance performance. The accompanying song is generally in adoration of the Supreme Being. Hence statement 2 is correct.**
- **After the shabdham, the dancer performs the varnam. The varnam which is the most important composition of the Bharatnatyam repertoire, it encompasses both nritta and nritya and epitomises the essence of this classical dance form. Hence statement 3 is correct.**
 - The dancer here performs complicated well-graded rhythmic patterns in two speeds showing the control over rhythm, and then goes on to depict in a variety of ways, through abhinaya the lines of the sahitya.
 - This portrays the dancer's excellence in abhinaya and also reflects the endless creativity of the choreographer. The varnam is by far one of the most beautiful compositions in Indian dance.
- **After the strenuous varnam, the dancer performs a number of abhinaya items expressing a variety of moods.** The bhava or rasa is woven into the sahitya and then expressed by the dancer. The common pieces are keertanam, kritis, padams and javalis. In the keertanam, the text is important whereas kriti is a composition in which the musical aspect is highlighted. Both are usually devotional in character and represent episodes from the lives of Rama, Siva, Vishnu, etc. Padams and javalis, are on the theme of love, often divine.
- **A Bharatnatyam performance ends with a tillana which has its origin in the tarana of Hindustani music.** It is a vibrant dance performed to the accompaniment of musical syllables with a few lines of sahitya. The finale of the piece is a series of well designed rhythmic lines reaching a climax. The performance ends with a mangalam invoking the blessings of the Gods.

Q 92.A

- **Statement 1 is correct:** During the reign of Akbar, painting was organized in one of the imperial establishments. Large number of painters from all the castes were invited to join. Daswant and Basawan were the famous painters. **Apart from illustrating Persian fables, painters were soon assigned the task of illustrating Persian text of Mahabharata, etc.** Thus, Indian scenes and themes came into vogue. **An illustrated manuscript of the Tuti-nama in the Cleveland Museum of Art (USA) appears to be the first work of the Mughal School.** The style of painting in this manuscript shows the Mughal style in its formative stage.
- **Statement 2 is correct:** Indian colors like peacock blue, the Indian red, etc. began to be used. **Above all the flat effect of the Persian Style began to be replaced by the roundness of Indian brush, giving pictures a three-dimensional effect.**
- **Statement 3 is not correct:** Under Akbar, **European painting was introduced at the court by the Portuguese priests.** Under their influence, the principles of foreshortening, whereby near and distant people and things could be placed in perspective was quietly adopted.
- **Statement 4 is not correct:** Mughal painting reached climax under the reign of Jahangir who had very discriminating eye. **It was a fashion in the Mughal school for the faces, bodies and feet of the people in a single picture to be painted by different artists.** Jahangir claims that he could distinguish the work of each artist in the picture. Apart from painting hunting, battle, and court scenes, under Jahangir special progress was made in portrait paintings and painting animals. It is of an high aesthetic merit. **It is primarily aristocratic and secular.** Following the example of the Mughal Emperor the courtiers and the provincial officers also patronised painting. They engaged artists trained in the Mughal technique of painting. **But the artists available to them were of inferior merit, those who could not seek employment in the Imperial Atelier which required only first-rate artists.**

Q 93.A

- **Ministry of Earth Sciences (MoES) through its autonomous Institute National Institute of Ocean Technology (NIOT) has developed Low Temperature Thermal Desalination (LTTD) technology for conversion of sea water to potable water which has been successfully demonstrated in Lakshadweep islands. Hence statement 1 is correct.**

- Till date, 4 LTTD plants have been successfully commissioned in the country, one each at Kavaratti, Minicoy, Agatti of Lakshadweep and one at Northern Chennai Thermal Power Station (NCTPS), Chennai.
- Based on the success of these plants, Ministry of Home Affairs (MHA) through Union Territory (UT) Lakshadweep has entrusted the work of establishing 6 more LTTD plants at Amini, Androth, Chetlet, Kadmat, Kalpeni and Kiltan with a capacity of 1.5 lakhs litres/day.
- **The LTTD technology is found suitable for Lakshadweep islands where the required temperature difference of about 15°C is found between sea surface water and deep-sea water. Hence statement 2 is not correct.**
- **The Reverse Osmosis, a membrane process, is globally accepted technology suitable for desalination of saline water, which is quite different from LTTD technology developed by the ministry. The LTTD is a process under which the warm surface sea water is flash evaporated at low pressure and the vapour is condensed with cold deep sea water. The LTTD technology does not require any chemical pre and post-treatment of seawater and thus the pollution problems are minimal and suitable for island territories.**
- The cost of desalination plant depends on a number of factors inter alia which includes technology used and location of plant. The total cost of the six LTTD plants in Lakshadweep islands is Rs. 187.75 cr.

Q 94.C

- **Statement 1 is not correct:** Balban stood forth as the champion of Turkish nobility. He refused to entertain for important government posts anyone who did not belong to a noble family. He excluded Indian Muslims from all positions of power and authority.
 - To emphasise that the nobles were not his equals, he insisted on the ceremony of sijada and paibos. These ceremonies were Iranian in origin and were considered un-Islamic.
- **Statement 2 is correct:** To keep himself well informed, Balban appointed spies in every department. He also organized a strong centralized army, both to deal with internal disturbances, and to repel the Mongols who had entrenched themselves in Punjab and posed a serious danger to the Delhi Sultanat.
- **Statement 3 is not correct:** Muhammad bin Tughlaq launched a scheme to extend and improve cultivation in the doab. He set up a separate department called diwan-i-amir-i-kohi. The area was divided into development blocs headed by an official whose job was to extend cultivation by giving loans to the cultivators and to induce them to cultivate superior crops—wheat in place of barley, sugarcane in place of wheat, grapes and dates in place of sugarcane, etc.
 - Balban organised a strong centralized army, both to deal with internal disturbances and to repel the Mongols. For this purpose, he reorganized the military department (diwan-i-arz) and pensioned off those soldiers who were no longer fit for service.

Q 95.C

- In a major milestone for China's space program, a new scientific lab (Wentian Space Station Module) docked with its under-construction space station, Tiangong, bringing it closer to completion. The Wentian module docked with Tiangong station's main Tianhe living module.
- **It is second of three modules to the China's permanent space station, Tiangong. First Tianhe module, the main living quarters, already placed in space in 2021 and last module Mengtian is expected to be launched in October 2022. Hence statement 1 is correct.**
- The Wentian lab module, 17.9 meters (59 feet) long, will be where astronauts can carry out scientific experiments, along with the other lab module yet to be launched - Mengtian ("Dreaming of the Heavens").
- **Wentian features an airlock cabin** that is to be the main exit-entry point for extravehicular activities when the station is completed.
- **It will also serve as short-term living quarters for astronauts during crew rotations on the station, designed for the long-term accommodation of just three astronauts. Hence statement 2 is correct.**
- The **Wentian is designed for science and biology experiments**, and the 23-ton lab module is heavier than any other single-module spacecraft currently in space, according to the state-owned Global Times.
- Meanwhile, Mengtian module is expected to be launched in October and, like Wentian, is to dock with Tianhe, forming a T-shaped structure. The completion of the structure, about a fifth of the International Space Station (ISS) by mass, is a source of pride among ordinary Chinese people.

Q 96.B

- **Statement 1 is not correct:** 'Latina' or the rekha-prasada type buildings have simple shikhara which is square at the base and whose walls curve or slope inward to a point on top. Another major type of architectural form in the nagara order is the phamsana. **Phamsana buildings tend to be broader and**

shorter than latina ones. Their roofs are composed of several slabs that gently rise to a single point over the centre of the building, unlike the latina ones which look like sharply rising tall towers.

- **Statement 2 is correct:** In many North Indian temples you will notice that the phamsana design is used for the mandapas while the main garbhagriha is housed in a latina building. Later on, the latina buildings grew complex, and instead of appearing like a single tall tower, the temple began to support many smaller towers, which were clustered together like rising mountain-peaks with the tallest one being in the centre, and this was the one which was always above the garbhagriha.
- **Statement 3 is correct:** These are rectangular buildings with a roof that rises into a vaulted chamber. The edge of this vaulted chamber is rounded, like the bamboo or wooden wagons that would have been drawn by bullocks in ancient times. They are usually called 'wagon- vaulted buildings'.

Q 97.D

- The institution of the **Iqta had been in force in the early Islamic world as a form of reward for services to the state.** It was **used in the Caliphate administration** as a way of **financing operations** and **paying civil and military officers.**
 - The Turkish (later known as Delhi Sultanate) conquests in the early 13th century displaced many local chiefs. To consolidate, the Turkish rulers made revenue assignments (iqta), instead of cash to their nobles (umara). **It was Iltutmish who institutionalized and popularized Iqta system in India. Hence statement 1 is correct.**
- **Assignments given to nobles carried administrative, military, and revenue collecting responsibilities.** Thus, the provincial administration was headed by the muqti/wali. They **had to maintain an army composed of horsemen and foot soldiers.** They have to furnish military assistance to the state when required. **Hence statement 3 is correct.**
- The **grant of iqta did not imply a right to the land nor was it hereditary,** however, **the holders of iqta tended to acquire hereditary rights in Firuz Tughlaq's reign.** These revenue assignments were transferable, the data-holder being transferred from one region to another every three or four years. **Hence statement 2 is correct.**

Q 98.D

- A geographical indication (GI) is a sign used on products that have a specific geographical origin and possess qualities or a reputation that are due to that origin. In order to function as a GI, a sign must identify a product as originating in a given place
 - Under Articles 1 (2) and 10 of the Paris Convention for the Protection of Industrial Property, geographical indications are covered as an element of IPRs. They are also covered under Articles 22 to 24 of the Trade-Related Aspects of Intellectual Property Rights (TRIPS) Agreement.
- In December 1999, the Parliament passed the Geographical Indications of Goods (Registration and Protection) Act, 1999. This Act seeks to provide for the registration and better protection of geographical indications relating to goods in India.
 - The Act is administered by the Controller General of Patents, Designs and Trade Marks- who is the Registrar of Geographical Indications. The Geographical Indications Registry is located at Chennai. The Act has come into force with effect from 15th September 2003.
- **Aranmula Kannadi (literally meaning Aranmula mirror) is a handmade metal-alloy mirror, unlike the usual glass mirror. Aranmula Kannadi is an extraordinary traditional handmade metal mirror which is famous for bringing prosperity, luck and wealth into life.**
 - What's unique about this front surface reflection mirror is that it eliminates secondary reflections that you typically see in back surface mirrors. Born out of Kerala's rich culture and its inclination towards metallurgical articles, the mirror is said to have tremendous spiritual value and brings good luck.
 - **In 2004, this rare craft belonging to a small town in Kerala was given worldwide recognition when a 45-centimeter tall Aranmula mirror was placed in London's British Museum and awarded a geographical indication (GI) tag.**
 - Owing to the unmatched splendor and religious significance of Aranmula Kannadi, Aranmula was declared a heritage village by Kerala Tourism in order to draw tourists' attention to this dying craft.
- **The making of Mysore Agarbathi became an organized industry in Bangalore during the 1900s.** The incense sticks are locally known as oodabathies (blowing fumes). Herbs, flowers, essential oils, barks, roots and charcoal are ground into a smooth paste, then rolled onto bamboo sticks and dried in the sun.
 - **The key ingredients Sandalwood and Ailanthus malabaricum (which yields halmadi) are indigenous to the forests of Karnataka, thus earning the product its GI tag status.**

- The Hmaram was and is a popular article of clothing for Mizoram women. It is a handmade shawl and is worn like a short skirt by being wrapped around the waist and tucked in on one side. It is worn with a white blouse for ceremonies, festivals, or special events. It is made of cotton and dyed with natural indigo. It enjoys a GI tag status.
- The Lock manufacturing industry in Dindigul is hundreds of years old. Dindigul locks can guarantee aspects like quality, full safety & key facilities, most importantly they are attractive and unique in design.
 - The popular kinds of locks that are manufactured in Dindigul are Mango lock, Door Lock, Trick Lock, Bell Lock, Drawer Lock, Shutter Lock & Book Shutter Lock.
 - Dindigul Locks are specifically purchased for Temples. Government institutions and business places because of its foolproof Safety.
 - Dindigul lock manufacturers claim that the precision lever mechanism in their locks is manually assembled with a unique key code and shackles are meticulous in design that gives greater resistance to anyone who tries to break in.
 - Dindigul locks enjoy GI tag status.
- Hence option (d) is the correct answer.

Q 99.B

- Two kinds of Mahajanapadas existed during the age of Buddha (6th century BC). They were the monarchies and non-monarchical (republics) states known as ganas or sanghas.
- The monarchies were governed by strong monarchs and had a centralised system of administration, the republics, on the other hand, **were ruled by the clan oligarchies i.e. by the dominant Kshatriya clans** residing in that territory.
- The administration system in republics was based on the collective participation of the clans in the decision-making.
- The **Mahajanpads of Magadha, Avanti, and Champa had a monarchical system** and were ruled by strong monarchs as the Mahajanapadas of Vajji, Mallas had a non-monarchical (republics) system.
- The **Vajjis represented a confederacy of eight clans** of whom the Videhas, Licchavis and the Jnatrikas were the most well known.
- In the neighborhood of Kosala **lay the republic clan of the Mallas**, whose territory touched the northern border of the Vajji state. One of the capitals of the Mallas lay at Kusinara, where Gautama Buddha passed away.
- Hence option (b) is the correct answer.

Q 100.D

- Recent context: The staff-level talks between Pakistan and the International Monetary Fund (IMF) concluded for the seventh and eighth review under **Extended Fund Facility (EFF)**. Hence option (d) is the correct answer.
- EFF was established to **provide assistance** to countries experiencing serious **payment imbalances** because of **structural impediments or slow growth** and an **inherently weak balance-of-payments position**.
- It **provides support for comprehensive programs** including the policies needed to correct structural imbalances over an extended period.
- It supports countries' economic programs aimed at moving toward a stable and sustainable macroeconomic position consistent with strong and durable poverty reduction and growth.
 - The ECF may also help catalyse additional foreign aid.
- **Eligibility** : It is available to all Poverty Reduction and Growth Trust (PRGT)-eligible member countries that face a protracted balance of payments problem.
- **Duration and repeated use**. Assistance under an ECF arrangement is provided for an initial duration from three to up to five years, with an overall maximum duration of five years.
 - Following the expiration, cancellation, or termination of an ECF arrangement, additional ECF arrangements may be approved.
- **Focused conditionality**: Under the ECF, member countries agree to implement a set of policies that will help them make progress toward a stable and sustainable macroeconomic position over the medium term.
 - These commitments, including specific conditions, are described in the country's letter of intent.